



Full length article

Relative population of $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ interstitial loops in alpha-Fe under irradiation: Effects of C15 cluster stability and loop one-dimensional movement

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ARTICLE INFO

Article history:

Received 22 October 2021

Revised 6 April 2022

Accepted 26 April 2022

Available online 29 April 2022

Keywords:

Irradiation

Iron

C15 cluster

Dislocation loop

1-d movement

ABSTRACT

The relative stability of $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ dislocation loops, originating from anisotropic elasticity of alpha-Fe, was extensively exploited to explain experimentally observed variation in the relative population of the two loop types under varying irradiation temperatures. In this study, using cluster dynamics simulations taking into account the coevolution of C15 Laves phase structure (cluster) and loop reaction caused by defect one-dimensional (1-d) movement, we reveal new mechanisms that may underpin the variations in loop population of different Burgers vectors. We identify that C15 clusters have twofold roles in mediating loop evolution: (1) acting as buffers of self-interstitial atoms, (2) initiating loop nuclei via their collapses. The latter plays a critical role in nucleation of $\langle 100 \rangle$ loops and subsequently affects loop evolution of both types. We show that the loop relative population undergoes further re-arrangement via a transfer reaction driven by loop long-range diffusion. Our findings essentially demonstrate that, apart from the loop stability, the relative population of two loop types is governed by at least two other factors: C15 cluster stability and loop 1-d movement.

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1. Introduction

Dislocation loops are typical irradiation defects formed in neutron-irradiated metallic materials [1–7]. They degrade the material properties by causing hardening and embrittlement of the irradiated materials [6,8–10]. Extensive studies employing experiments [2–4] and simulations [11–13] investigated their formation and growth in various metallic materials. Particularly the study on loop evolution in iron and iron-based steels attracts tremendous attention. This is largely due to the fact that iron-based steels have extensive applications in nuclear structure components for current operating reactors, and are also regarded as candidate structure components for future fusion reactors [14–16]. Moreover, for iron itself, due to particular magnetic property, dislocation loops accommodated in its system present different features compared with those in other body-centered cubic (bcc) metals [17]. For instance, loop formation with Burgers vector $\langle 100 \rangle$, which is rarely observed in the non-magnetic bcc metals.

The frequent occurrence of the $\langle 100 \rangle$ -type prismatic edge dislocation loops was first observed in iron irradiated at 500–600 °C by Masters [18,19]. In iron-based Eurofer97 steel, Klimenkov et al. [20] also observed the dominant occurrence of the $\langle 100 \rangle$, as op-

posed to the $1/2\langle 111 \rangle$, Burgers vector configurations of prismatic dislocation loops after neutron irradiation at temperatures over 350 °C. Theoretically, Marian et al. [12] and Xu et al. [13] predicted that $\langle 100 \rangle$ loops can form via the interaction of two $1/2\langle 111 \rangle$ loops. However, this might not be the reason, at least not the governing factor for the formation of $\langle 100 \rangle$ dislocation segments observed in iron. This is because the proposed mechanism requires the collision between two migrating $1/2\langle 111 \rangle$ loops with similar sizes and overlapping glide cylinders, making it unlikely in reality. Furthermore, small prismatic $1/2\langle 111 \rangle$ loops in non-magnetic bcc metals are just as mobile as in bcc iron, but $\langle 100 \rangle$ loops are not observed. To address this anomaly, Dudarev et al. [21] calculated the anisotropic self-energies of $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ dislocations in bcc iron, and thus derived the relative stability of prismatic $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ dislocation loops. A gradual transition in loop types with increasing temperature was predicted. Later Yao et al. [22] using heavy-ion irradiation on high-purity iron observed a compatible picture that $\langle 100 \rangle$ loops become more energetically favored with increasing temperature. However, the mechanism of the loop type transition is still not clear. $1/2\langle 111 \rangle$ loops transforming directly into $\langle 100 \rangle$ loops were not observed in the experiments. Instead, the abrupt disappearance of $1/2\langle 111 \rangle$ loops occurred at temperatures over 465 °C. These experimental observations, therefore, cannot justify the mechanism for the origin of $\langle 100 \rangle$ loops in irradiated bcc iron.

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Recently, an increasing number of studies using molecular dynamics (MD) simulations reported the formation of a three-dimensional C15 Laves phase structure in bcc iron [23–28]. The C15 structures were predicted to have lower energies than their loop counterparts at small sizes [23–28]. This indicates that they are energetically more stable than loops within a certain size range. When C15 clusters grow beyond a certain size range via absorbing mobile self-interstitial atoms (SIAs), the relative stabilities are reversed. They become unstable and collapse into loops. Both loop types are possible collapse products [25–27]. This, thus, essentially provides a mechanism for the formation of $\langle 100 \rangle$ loops. If assuming some percentages of loops have an early embryo originating from C15 clusters, the ratio of different-type loop embryos formed via the collapse would have an impact on their relative population. The temperature dependence of such a ratio may help explain the experimentally observed dominant occurrence of $\langle 100 \rangle$ loops at high temperatures. Unfortunately, dedicated reports on the temperature dependence of C15 cluster stability are not available so far. Moreover, apart from the collapse of C15 clusters, $1/2\langle 111 \rangle$ loops can directly nucleate from in-cascade clustering and via the accumulation of migrating SIA (dumbbells) [29–31]. Even assuming that C15 clusters collapse preferentially into $\langle 100 \rangle$ loops at high temperatures, a large number of $1/2\langle 111 \rangle$ loop nuclei are still continuously generated via cascades and mobile SIAs. Therefore, sole consideration of the formation and collapse of C15 clusters cannot fully explain the dominance of $\langle 100 \rangle$ loops at high temperatures.

In addition to the collapse of C15 clusters, MD simulations also proposed the formation of $\langle 100 \rangle$ loops by the cascade overlapping with pre-existing SIA clusters [32,33]. The mechanism relies on the effects of the molten cascade core reforming parts of pre-existing SIA clusters (dislocation loops and C15 clusters). A similar mechanism was reported early by Terentyev et al. [34], who found that small loops can be dissolved by a cascade and then be reformed to loops with different Burgers vectors. In Fe, the transformation from C15 clusters and $1/2\langle 111 \rangle$ loops into $\langle 100 \rangle$ loops by cascades was observed with a probability of 10%–20% at full overlap [33]. It can be expected that such overlapping events become increasingly likely at high dose (rate) conditions, typical for materials in nuclear reactors and ion irradiation experiments. Though this process provides an additional nucleation source for $\langle 100 \rangle$ loops, the effects are mainly on small SIA clusters. To explain the experimentally observed dominance of loops with a particular type, it is required to consider the kinetics involved in loop long-term evolutions, especially the reaction between loops provoked by loop 1-d movement.

Using *in-situ* TEM examinations, Arakawa et al. [35] demonstrated a loop reaction that followed the collision between two loops with different Burgers vectors in bcc iron. The result of the reaction was the absorption of the smaller loop by the larger one. This essentially provides a mechanism for the loop transfer between different Burgers vectors. The reaction requires collision between two loops, indicating that the mobile loops ultimately determine the reaction occurrence. It is known that the prismatic $1/2\langle 111 \rangle$ loops perform the 1-d movement in bcc iron even in the absence of stresses [36]. When $1/2\langle 111 \rangle$ loops migrate towards and collide with $\langle 100 \rangle$ loops, they either absorb $\langle 100 \rangle$ loops become large or disappear as being subsumed by $\langle 100 \rangle$ loops. Reaction results depend on relative loop sizes. Such a mechanism is expected to re-arrange the loop populations during the long-range diffusion of mobile loops.

It seems that the unsolved issue – the variation in the relative population of two loop types under varying irradiation temperatures – can be rationalized by considering the above-mentioned kinetics occurring during irradiation, i.e., the cascade overlap with pre-existing SIA clusters, C15 cluster stability and loop reactions driven by defect long-range diffusion. The first two terms can act

as sources for $\langle 100 \rangle$ loop nucleations, and the latter modifies the population of both $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ loops during their long-term evolution. To test this idea, in this study, we developed a cluster dynamics model considering the above kinetics. For instance, the cascade reforming small C15 clusters and dislocation loops into different types of SIA clusters was implemented in the model. The model also incorporated the formation of C15 clusters and their subsequent collapse into loops. Moreover, the model considered the one-dimensional reaction kinetics in the case where 1-d diffusing loops are involved, following the description of trap-mediated loop migration proposed by Kohnert and Wirth [37]. The assumption of spherical reaction volumes [38–41], which have been widely accepted in prior models becomes invalid for the 2-dimensional loops with sizes over a few nanometers. In this model, the physically meaningful annular cross sections were adopted to characterize the interaction between loops, aiming to account for the repulsion caused by strain fields of coaxial loops [42]. The experimental data [22,43,55] obtained using *ex-situ* and *in-situ* TEM characterization for irradiated bcc iron was employed to validate the model. Effects of the cascade overlap with pre-existing SIA clusters, C15 cluster stability and defect long-range diffusion on the development of both loop types were evaluated.

2. Model description

2.1. Basic formalism

The cluster dynamics (CD) model was built based on classical rate theory. The population of each object evolving in the model was quantified by integrating a coupled set of diffusion-assisted reaction equations. For simulating metallic materials under irradiation, point defects like SIAs, vacancies, and their aggregational structures are usual objects traced in a modeling system [40,41,44,45]. The governing equation for each object can be written in a general form as

$$\begin{aligned} \frac{dC(\alpha)}{dt} = & G_{\text{irra}}^{\alpha} - L_{\text{sink}}^{\alpha} - \sum_{\beta} k_{\alpha}^{\beta} C(\alpha) C(\beta) - \sum_{\beta} g_{\alpha}^{\beta} C(\alpha) \\ & + \sum_{\alpha', \beta} k_{\alpha'}^{\beta} C(\alpha') C(\beta) + \sum_{\alpha'', \beta} g_{\alpha''}^{\beta} C(\alpha'') \end{aligned} \quad (1)$$

where G_{irra}^{α} and L_{sink}^{α} represent the rate of the defect α generated from irradiation and the rate of its loss to sinks (grain boundaries, dislocations, etc.), respectively. k_{α}^{β} and g_{α}^{β} are respectively the rate of the defect α absorbing and emitting the mobile defect β . $C(\alpha)$ and $C(\beta)$ denote defect concentrations. The last two terms describe the generation of defects α via the reaction of their configuration-nearby defects (α' and α'') with defects β .

In this study, the defects in the simulation include SIAs, C15 clusters, self-interstitial dislocation loops of $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ types, vacancy and vacancy clusters (voids). Vacancy-type dislocation loops are not considered. Mobile defects include SIAs, $1/2\langle 111 \rangle$ loops of all sizes, vacancy and vacancy clusters containing no more than 5 vacancies. SIA clusters containing 3 or fewer SIAs perform 3-d movement, while larger SIA clusters constructed as $1/2\langle 111 \rangle$ loops perform 1-d movement, or are constructed as sessile C15 clusters. Vacancy-type defects if mobile also perform 3-d movement. The mobility of defects implemented in the model was shown in Table 1. The model also considers the irradiation-induced random walk of dislocation loops. This athermal diffusivity originates from irradiation-induced hops of loops between trapping sites. We adopted the analytical expression given in Ref. [37].

$$D_i^{dt} = \nu_i^{dt} \frac{\lambda_i^2}{2N} \quad (2)$$

Table 1

Mobility parameters of point defects and defect clusters as implemented in the model.

Cluster	Pre-exponential factor	Migration energy (eV)
I_1 (3-d)	$D_0(I_1) : 10^{13} a_0^2$ (a_0 the lattice parameter)	0.34 *
I_2 (3-d)	$D_0(I_1)/2$	0.42 *
I_3 (3-d)	$D_0(I_1)/3$	0.43 *
$I_4 - I_\infty$ (free, 1-d)	$D_0(I_1)/n^{0.7}$ (n the number of SIA contained in $1/2<111>$ loop) †	0.10 †
$I_4 - I_\infty$ (trapped, 1-d)	$D_0(I_1)/n^{0.7}$ (n the number of SIA contained in $1/2<111>$ loop) †	$4 \leq n \leq 10, 0.55;$ $10 < n \leq 37, 1.0;$ $n > 37, 1.3$ †
V_1 (3-d)	$D_0(V_1) : 10^{13} a_0^2$	1.25 ‡
V_2 (3-d)	$D_0(V_1)/2$	1.3 [42]
V_3 (3-d)	$D_0(V_1)/3$	1.3 [42]
V_4 (3-d)	$D_0(V_1)/4$	1.3 [42]
$V_5 - V_\infty$	0	-

* Migration energies of small SIA clusters adopted the *ab-initio* values calculated by Fu et al. [46].

† Pre-exponential factor and migration energy of large SIA clusters (mobile loops) adopted the values given by Kohnert and Wirth [42]. This study allows higher mobility for small clusters (≤ 37 SIAs). A similar setup for loop migration was proposed in Ref. [37].

‡ The effective vacancy migration energy in iron containing C, N impurities is much higher than the *ab-initio* value (~ 0.67 eV) in ideal pure iron [46,47].

where ν_i^{dt} is hop frequency of the loop, λ_i is the discrete hop length, which is associated with jumps between trapping sites. $\lambda_i = (4\pi r_i r_t \rho_t)^{-1}$, where r_i is loop radius, r_t is the interaction distance between loop and trap, and ρ_t is trap density. The hop frequency ν_i^{dt} depends on dose rates, and a hop activation value of 30/dpa was proposed for ion irradiation in Ref. [37]. N is the overall dimensionality of the hops and is assumed to be three dimensional ($N = 3$).

The absorption coefficient of reaction between 3-d diffusers and 3-d structures, like C15 clusters absorbing SIAs, can be calculated using the conventional form [38,39]

$$k_\alpha^\beta = 4\pi (D_\beta + D_\alpha) (r_\beta + r_\alpha), \quad (3)$$

where D and r indicate the diffusivity and radius of the reactants, respectively. For the reaction between 3-d diffusers and stationary loops, where non-spherical reaction volumes involve, the reaction rate was estimated following Kohnert and Wirth's approach (see Eq. (11) in Ref. [42]). For the reaction caused by the 1-d diffusers, the reaction rate adopted the widely accepted solution

$$k_\alpha^\beta = \frac{2D_\alpha \sigma_{\alpha,\beta}}{\lambda_\alpha}, \quad (4)$$

where $\sigma_{\alpha,\beta}$ is the reaction cross-section, the formulas of which for 1-d diffusers approaching various types of interaction partners were given in Ref. [42]. λ_α is the mean free path of the 1-d diffuser, which can be computed by $\lambda_\alpha^{-1} = \sum_\omega \sigma_{\alpha,\omega} C(\omega)$, where ω indicates all sinks and defects interacting with the 1-d diffuser. Therefore the magnitude of reaction rates is dynamic and depends on the varying defect population. The emission coefficient g_α^β is related to the absorption coefficient of daughter defects and the binding energy $E_b^\beta(\alpha)$ of the parent α . Only point defect emission from clusters is allowed in the model.

$$g_\alpha^\beta = \frac{1}{\Omega} k_{(\alpha-\beta)}^\beta \exp\left(\frac{-E_b^\beta(\alpha)}{k_B T}\right) \quad (5)$$

The binding energy of point defects (SIA, vacancy) to SIA and vacancy clusters was calculated, respectively, by:

$$E_b^{SIA}(n) = \begin{cases} E_f^{SIA} + E_f(n-1) - E_f(n) & \text{(for SIA clusters)} \\ E_f^{SIA} + E_f(n+1) - E_f(n) & \text{(for vacancy clusters)} \end{cases} \quad (6.1)$$

$$E_b^V(n) = \begin{cases} E_f^V + E_f(n+1) - E_f(n) & \text{(for SIA clusters)} \\ E_f^V + E_f(n-1) - E_f(n) & \text{(for vacancy clusters)} \end{cases} \quad (6.2)$$

where E_f^{SIA} , E_f^V are the formation energy of SIA and vacancy in iron, respectively. $E_f(n)$ is formation energy of SIA clusters containing n SIAs, or of voids containing n vacancies. The formation energy of voids was calculated using $E_f^{Void}(n) = 4\pi (r_V^0)^2 \gamma$, with r_V^0

defined as $\frac{4}{3}\pi (r_V^0)^3 = n \frac{a_0^3}{2}$, where γ and a_0 are the surface energy and lattice parameter, respectively. No temperature dependence for void energetics was considered. For SIA clusters, the formation energy of loops with temperature dependence was calculated under the framework of anisotropic elasticity in alpha-Fe [48]. The formation energies of $<100>$ and $1/2<111>$ loops were calculated in Section 2.2.

The inherent sinks I_{sink}^α considered in the model includes dislocation lines (DL), grain boundaries (GB) and free surfaces (surf). The corresponding sink strengths, $L_{sink}(DL)$, $L_{sink}(GB)$ for 1-d and 3-d diffusers are different. The model adopts the expressions given in Ref. [49]

$$L_{sink}(DL) = \begin{cases} 8r_{dc}\rho_{dl}^{3/2} & \text{(for 1-d diffusers)} \\ \rho_{dl} & \text{(for 3-d diffusers)} \end{cases} \quad (7.1)$$

$$L_{sink}(GB) = \begin{cases} 24/d^2 & \text{(for 1-d diffusers)} \\ 6/d^2 & \text{(for 3-d diffusers)} \end{cases} \quad (7.2)$$

where ρ_{dl} is dislocation density, r_{dc} the dislocation capture radius, d the grain size. Additionally, a bias factor Z was applied to $L_{sink}^{SIA}(DL)$, i.e., loss of single SIAs to dislocations. For *in-situ* experiments using thin foils, free surfaces are strong sinks [49]

$$L_{sink}(surf) = \begin{cases} 8\cos^2\varphi/l^2 & \text{(for 1-d diffusers)} \\ 2/l^2 & \text{(for 3-d diffusers)} \end{cases} \quad (7.3)$$

where l is the foil thickness, φ the angle between loop mobility trajectory and surface normal.

Irradiation cascades directly generate SIA and vacancy clusters in the material system. To mimic the primary state of damage under ion irradiation, Marinica performed a series of cascade simulations with a kinetic energy of 20 keV for the primary knock-on atom and various initial orientations at temperatures ranging from 300 to 900 K [23]. It was found that about half of SIAs were clustered. In this study, the fraction of SIA (vacancy) in clusters of size n generated via the in-cascade clustering adopted the exponential expression [50]

$$f_{irra}^{SIA}(n) = 0.89e^{-0.67n} \quad (8.1)$$

$$f_{irra}^V(n) = 2.93e^{-1.5n} \quad (8.2)$$

where the parameters for SIA clusters $f_{irra}^{SIA}(n)$ and vacancy clusters $f_{irra}^V(n)$ were obtained via fitting the results of cascade simulations using energies up to 200 keV. This expression reproduces that about half of cascade-induced SIAs ($\sim 54\%$) are accommodated in clusters. The production rate of n -size clusters, i.e., C_{rra}^α in Eq. (1), then can be calculated to be $\frac{\eta G_{dpa}}{V_{at}} \cdot \frac{f_{irra}^\alpha(n)}{n}$, where η , G_{dpa} , V_{at} and n are the damage efficiency, dpa rate, atomic volume and number of SIAs constituting the cluster, respectively.

Among the cascade-induced SIA clusters, 5% of SIA clusters were C15 type with sizes up to 12 SIAs, as reported by Marinica using the M07 potential [23]. A higher percentage (5%–10%) of C15 clusters was found by Byggmästar using the M07-B potential [27]. In the current model, we set that 7% of the cascade-induced SIA clusters in the size range of 4 to 12 SIAs are C15-type. The rest of SIA clusters larger than 4 SIAs are treated as $1/2<111>$ loops, and

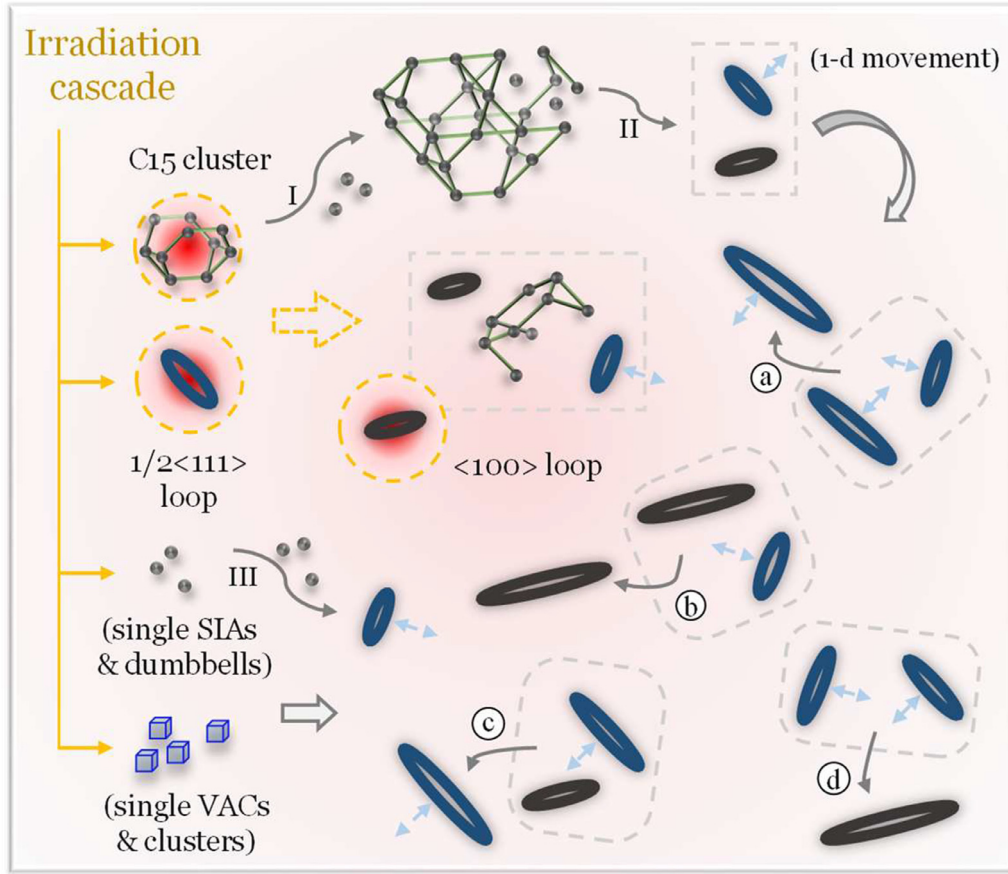


Fig. 1. Schematic drawing of the possible kinetics involved in the evolution of dislocation loops during irradiation. The circle-dashed areas indicate the cascade overlap with pre-existing SIA clusters, which were produced via in-cascade clustering or accumulation of migrating SIAs (Process III). Processes I–II indicate the growth of C15 clusters via absorbing migrating SIA (dumbbells), and the collapse of C15 clusters into loops, respectively. Process III depicts the nucleation of $1/2\langle 111 \rangle$ loops via the accumulation of migrating SIA (dumbbells). Processes a–d illustrate four modes of loop coalescence caused by loop 1-d diffusion. The type of resultant loops follows the larger loop participating in the reaction except the Process d, in which a $\langle 100 \rangle$ loop forms by the reaction of two $1/2\langle 111 \rangle$ loops of the same size.

clusters containing 2 or 3 SIAs are regarded as 3-d diffusing SIA dumbbells [46].

Apart from the direct loop formation ($1/2\langle 111 \rangle$ type) in cascades, the model follows the prediction of MD simulations that both $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ loops can nucleate from the collapse of C15 clusters [23–27]. Fig. 1 shows the possible kinetics involved in the evolution of dislocation loops during irradiation. When C15 clusters grow to critical sizes (Process I in Fig. 1), they collapse into loops of both types (Process II in Fig. 1). The fraction of the transformation into $\langle 100 \rangle$ loops determines the nucleation of $\langle 100 \rangle$ loops. Unfortunately, currently there is no conclusive data about such fractions, though several studies have reported some sparse trends about it [25–27]. This parameter was set adjustable based on available simulation results.

The model also considers the transformation between SIA clusters of different structures via cascades overlapping with pre-existing SIA clusters [32,33]. Only the full overlaps between the cascade and SIA clusters smaller than 50 SIAs are taken into account. The process was also schematically shown in Fig. 1. Such kinetics are implemented in the model by adding corresponding transformation terms to the governing equation Eq. (1). For instance, the governing equation of $\langle 100 \rangle$ loops is updated as

$$\frac{dC_{100}(n)}{dt} = \xi + f_{111to100}(n)F_{ol}C_{111}(n) + f_{C15to100}(n)F_{ol}C_{15}(n) - (f_{100to111}(n) + f_{100toC15}(n))F_{ol}C_{100}(n) \quad (9)$$

where $C_{100}(n)$, $C_{15}(n)$, $C_{111}(n)$ are the concentration of $\langle 100 \rangle$ loops, C15 clusters and $1/2\langle 111 \rangle$ loops containing n SIAs, respec-

tively. ξ represents all the original terms on the right-hand side of Eq. (1). $f_{111to100}(n)$ denotes the probability of $1/2\langle 111 \rangle$ loops being reformed to a $\langle 100 \rangle$ loop by an overlap event with cascade. $f_{C15to100}(n)$, $f_{100to111}(n)$ and $f_{100toC15}(n)$ have similar meanings. F_{ol} is the probability of a SIA cluster fully overlapping the molten region of cascade in unit time. This quantity essentially describes the rate of full overlap events that happened to a SIA cluster during irradiation. Governing equations for C15 clusters and $1/2\langle 111 \rangle$ loops have a form similar to Eq. (9).

It is worth noting that when a cascade overlaps with SIA clusters, its defect production reduces. A cascade when fully overlapped with SIA clusters generates zero Frenkel pairs [33]. Additionally, a SIA cluster after experiencing an overlapping event becomes smaller than its original size [33]. These effects are ignored in the model, considering that (1) the number of fully overlapping cascade events, compared to all cascade events, is expected to be negligible; (2) the “lost” SIAs of the original clusters by a cascade might easily diffuse to (be absorbed by) the SIA clusters after the transformation in the cascade region [33].

As $1/2\langle 111 \rangle$ loops have four possible variants, the cross-section of reaction between them varies with their relative orientations. For instance, the reaction cross-section between aligned pairs is larger than the unaligned ones. As the current model does not distinguish loop variants, we proposed a nominal cross-section for the reaction between $1/2\langle 111 \rangle$ loops. If each loop variant shares $1/4$ of the total concentration, the ratio of the possibility of two random variants being aligned and unaligned is $\sim 1 : 3$ ((c-1): 3c,

where c is the number of each variant). The nominal cross-section, thus, can be computed by weighting these possibilities to corresponding cross-sections of the aligned and unaligned reactions.

Loop coalescence occurs when a diffusing loop encounters another (a mobile or immobile loop). Fig. 1 (Process a–c) shows three modes of coalescence reactions considered in the simulation. One principle was held that the bigger loop absorbs the smaller one, aiming to be consistent with the experimental findings by Arakawa et al. [35]. The reaction provides a possibility for the transfer between $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ loops. However, the kinetics is different from that of two $1/2\langle 111 \rangle$ loops encountering and then transforming into a $\langle 100 \rangle$ loop [12]. Such a formation mechanism is schematically indicated by the Process d in Fig. 1. The current model includes this reaction. For four types of $1/2\langle 111 \rangle$ variants containing an identical number of SIAs, the possibility of two random variants encountering and forming a $\langle 100 \rangle$ loop is $\sim 3/8$ ($3c/(8c-2)$, c the number of each variant). The possibility of $3/8$ seems to be rather high, however, it is valid only for the reactions between loop variations of the same size. The rest of $\sim 5/8$ percent of the reactions between $1/2\langle 111 \rangle$ loops of the same size give rise to larger $1/2\langle 111 \rangle$ loops, similar to the Process a in Fig. 1. Literature also reported the transition from a single $1/2\langle 111 \rangle$ loop to a $\langle 100 \rangle$ loop [51,52], however, the mechanism is still not entirely clear. Therefore this reaction is not considered.

It is worth noting that, according to MD simulations [11,53,54], interstitial clusters diffuse along 1-d trajectories with very low activation energies (less than 0.1 eV) in pure iron. However, *in-situ* experiments showed a different scene where mobile loops performed rather limited movements even in high-purity iron [22,43,55]. Such inconsistency between expected and observed behaviors has been reconciled by “traps”, which bind to dislocation loops and lower their mobilities [37]. The trap was defined as a variety of impurity or impurity defect complex [37]. The mechanism that impurities lower the mobility of interstitial loops can be found in Ref. [56]. The current model considers $1/2\langle 111 \rangle$ loops in two states: free and trapped, as shown in Table 1. Free $1/2\langle 111 \rangle$ loops after nucleation can be trapped when encountering traps during their movements.

2.2. Loop energetics estimated by anisotropic elasticity

Estimation of loop binding energies is a premise to calculate the loop emission rates. However, assessing the free formation energy of a large loop containing up to tens of thousands of SIAs is not an easy task for conventional MD or density-functional theory (DFT) calculations. Here, the formation energy of prismatic loops was estimated using anisotropic elasticity theory [21,48]. Loops were assumed to have rhombus shapes, and the total free energy is calculated as the sum of anisotropic elastic self-energies and corresponding core-energies of edge dislocations constituting the loop. The temperature dependence of loop energy was also addressed by including the temperature dependence of elastic constants in bcc iron [48].

Previous studies have shown that $\langle 100 \rangle$ loops with sides parallel to $[100]$ direction have the lowest energy [48]. Fig. 2(a) shows the formation energy of $\langle 001 \rangle[100]$ loops as a function of loop size. One can see that loop formation energy decreases with temperature. Formation energy of $1/2\langle 111 \rangle$ loops with sides parallel to $[11\bar{2}]$ and $[1\bar{1}0]$ directions was also calculated. The comparison of energy between loops of two types was shown in Fig. 2(b). Generally, $\langle 111 \rangle[11\bar{2}]$ loops have lower energies than $\langle 111 \rangle[1\bar{1}0]$ and $\langle 001 \rangle[100]$ loops at large sizes. The cross-overs in energy between $\langle 111 \rangle[11\bar{2}]$ and $\langle 001 \rangle[100]$ loops increase from ~ 55 SIAs at 300°C to ~ 160 SIAs at 500°C . In the current model, $\langle 100 \rangle$ and $1/2\langle 111 \rangle$ loops were considered to have sides parallel to $[100]$ and $[11\bar{2}]$ directions, respectively.

2.3. Grouping scheme on loop population

To model large loops, for instance $\langle 100 \rangle$ loops with 40 nm in size, i.e., containing 31,116 SIAs, a large number of coupled differential equations have to be solved. To reduce computation time, the grouping method proposed by Golubov et al. (G-grouping) [58], which uses two equations per cluster group to ensure the conservation of cluster density and the conservation of mass, is implemented in the model. The G-grouping uses two moments (L^0 and L^1) to define the cluster size distribution within the group. For the j th group containing clusters of sizes $x_j^0 - \Delta x$ to $x_j^0 + \Delta x$, the cluster density of size x can be defined using the size distribution function (SDF) [59]

$$C_j(x) = L_j^0 + L_j^1 \cdot \frac{x - x_j^0}{1 + \Delta x_j} \quad (10)$$

where x_j^0 is the center cluster size of the group, $2\Delta x_j + 1$ represents the group width. The number density of clusters in the j th group can be accurately represented by L_j^0 , while adjustment of L_j^1 allow precise conservation of mass in the group. Any density change to individual clusters in the group alters the two moments of the group, following the two equations

$$\frac{dL_j^0}{dt} = \frac{1}{1 + 2\Delta x_j} \sum_{x=x_j^0-\Delta x}^{x_j^0+\Delta x} R_j(x) \quad (11.1)$$

$$\frac{dL_j^1}{dt} = \frac{3}{\Delta x_j(1 + 2\Delta x_j)} \left[\sum_{x=x_j^0-\Delta x}^{x_j^0+\Delta x} R_j(x) \cdot (x - x_j^0) \right] \quad (11.2)$$

where $R_j(x) = \frac{dC_j(x)}{dt}$ is the change in number density of cluster x in the j th group, which is already defined by the general governing equation Eq. (1). Using the cluster SDF defined by Eq. (10) to replace the cluster concentration in Eq. (1), the governing equations for groups, Eqs. (11), becomes the function of L^0 and L^1 of relevant groups. For simplicity, the detailed expressions are not listed here. The interested readers can refer to Ref. [59] for details.

The moment grouping scheme generates more terms than the conventional rate equations. For reactions between two groups, each absorption term k in the conventional rate equations is expanded to four k' for L_j^0 and L_j^1 , i.e., totally eight absorption terms generate for the moment scheme. For each emission term g , the moment scheme generates four times emission terms g' . Enforcing constant cluster properties across each group allows the group reaction rates k' and g' to be converted from k and g (of the center cluster of the group), respectively. The converting factors depend on the grouping setup only, and can be calculated by analytical expressions [59]. Therefore, they need to be computed only once, before time integration begins.

As described in the above sections, the model considers rather complex reactions for small loops, like nucleation from cascades, nucleation from the collapse of C15 clusters, and overlapping with cascades. This loop population is not grouped. The grouping is set for loops larger than 60 SIAs. In the model, we set 45 groups for large $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ loops. The first 40 groups, with a fixed width of 1.0 nm (except the first one), cover the loop size range of interest in this study. Though the group width is identical, the span of cluster size in terms of the number of SIAs within each group increases with the group index (Fig. 3). The last 5 groups have larger non-identical widths, leading to the largest loops containing 183,955 SIAs, corresponding to the largest $1/2\langle 111 \rangle$ loop of 105 nm and $\langle 100 \rangle$ loop of 97 nm.

Though the grouping scheme is used, directly integrating the ordinary differential equations (ODE) is still quite time-consuming.

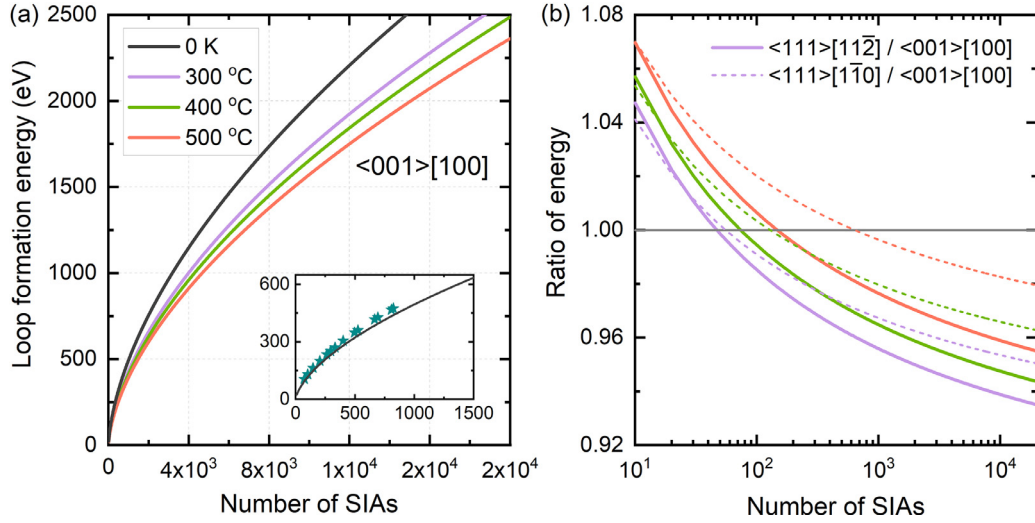


Fig. 2. Loop formation energy estimated by the anisotropic elasticity theory. (a) Formation energy of rhombus $\langle 001 \rangle$ loops with sides parallel to $[100]$ direction. Inset shows the comparison of energy with the ones calculated by the interatomic magnetic potential at 0 K [57]. (b) Ratio of the formation energy of $1/2 \langle 111 \rangle$ and $\langle 001 \rangle$ prismatic loops, plotted as a function of loop size. The energies of $1/2 \langle 111 \rangle$ loops with sides parallel to $[11\bar{2}]$ (solid lines) and $[1\bar{1}0]$ (dashed lines) directions are estimated respectively to compare with those of $\langle 001 \rangle$ loops with sides parallel to $[100]$ direction.

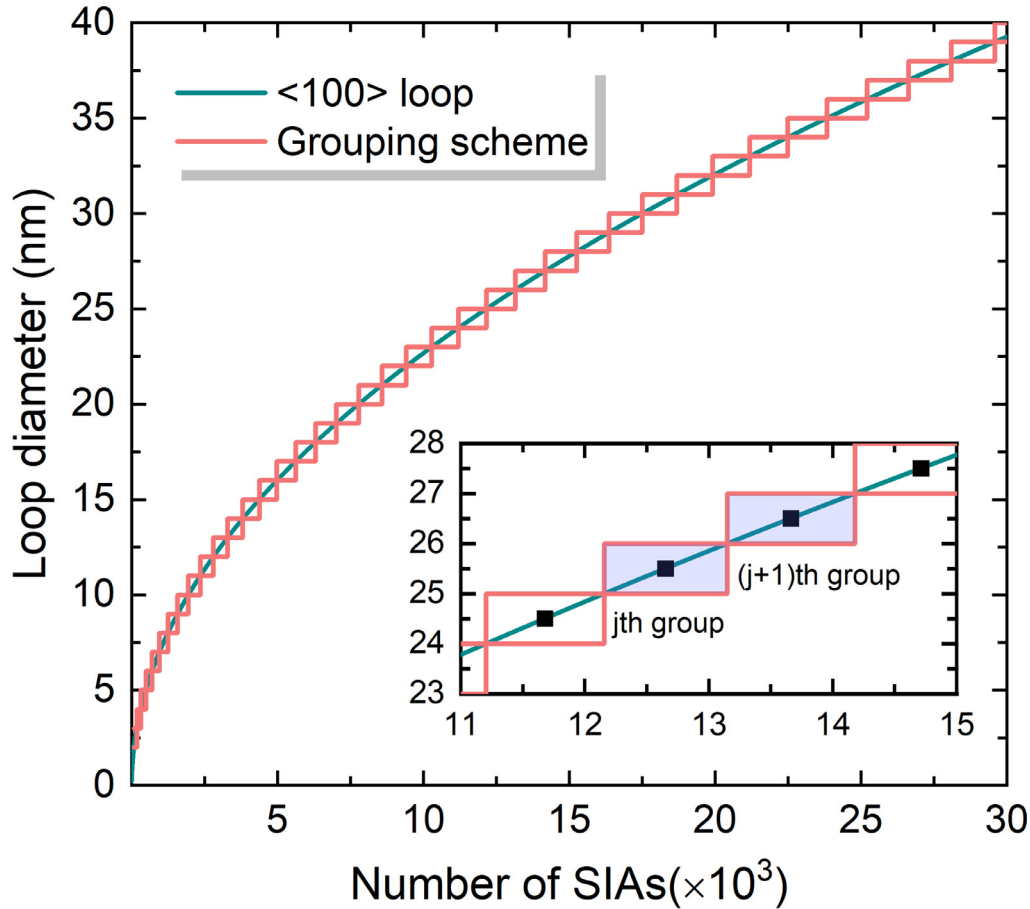


Fig. 3. Grouping scheme implemented in the cluster dynamic simulations. Individual loop diameter as a function of the number of SIA, and grouping setup for loops of different sizes. Inset shows a magnified view of the group setup and center clusters (square points) of groups.

Thus, the model defines the Jacobian matrix of the ODE system using the form $J_{ij} = \partial(\frac{dC_i}{dt}) / \partial C_j$, where C_i and C_j are the density of clusters (or moments L^0 and L^1 of cluster groups), respectively. This matrix is sparse, as defect clusters are coupled with each other through mobile defects of limited types.

The numerical solution of the Jacobian matrix is calculated using the python wrappers of odespy package [60] based on the odepack algorithms [61]. The computation time to achieve a physical time equal to $t = 2500$ s is 485 s on a 3.19 GHz processor.

3. Results

3.1. Validation of the CD model

3.1.1. Experimental conditions used in simulations

Model validation employs the experimental size distribution of dislocation loops probed by TEM in irradiated pure iron [22,55]. Bulk samples and thin foils were irradiated by 150 keV Fe⁺ ions with nominal dose rates (in peak damage regions) of 4×10^{15} ions · m⁻²s⁻¹ at 300 °C and 8×10^{14} ions · m⁻²s⁻¹ at 400 °C [22,43,55]. *In-situ* TEM characterization at 400 °C was done near the peak damage areas (at depth of ~ 28 nm), where the dpa rate is 5.2×10^{-4} dpa/s. *Ex-situ* TEM characterization was performed at depths of 60–100 nm for samples irradiated at 300 °C, leading to an effective dpa rate of 7.0×10^{-4} dpa/s calculated using the SRIM software [62]. The experimental details can be found in Refs. [22,43,55]. Input parameters related to irradiation conditions are listed in Table 2, along with key material parameters.

3.1.2. Parameters for cascade overlap with SIA clusters

As depicted in Fig. 1 and quantified by Eq. (9), cascades transform pre-existing SIA clusters into C15 clusters and loops of both Burgers vectors. However, two additional cluster types with non-negligible probabilities were also present after the overlap: (1) mixed structures containing segments of both C15 cluster and loop, and (2) clusters with “None” structure which cannot be identified as any specific type [33]. For the mixed structures, annealing tests showed that some of them were unstable and re-arranged into dislocation loops of either type, while the remained ones were quite stable even at 1000 K [33]. Cascades overlapping with the mixed structures was also possible and it converted the structures mainly into $1/2\langle 111 \rangle$ loops [32]. For the “None” structure clusters, there is no detailed information about their stability. However, in an early MD simulation work by Gao [29], it was found that a high percentage (25–35%) of sessile clusters were formed during the quenching stage of a cascade. Most sessile clusters transform into glissile clusters at about 500 K, while some of them are reformed to a compact three-dimensional configuration of high symmetry consistent with the C15 structure. Note that the sessile clusters

mentioned here were not $\langle 100 \rangle$ loops, as very rare simulation reported $\langle 100 \rangle$ loop formation by single cascades [66]. It is speculated that these sessile clusters were the “None” structure clusters, similar to those caused by the cascade overlapping with SIA clusters. A further speculation is that the “None” structure clusters resulting from the overlaps between cascade and SIA clusters are also instable, and they are likely to be transformed into loops and C15 clusters. We believe that cascades overlapping with the “None” structures is also possible and reform them into stable structures, like C15 clusters or loops of both types.

To directly describe the mixed and “None” structures in the current model is not possible for lack of information about their energetics, configurations and importantly the interaction with other defects. Their role, however, is represented by the possible stable forms they may transform into. In the model, the percentage of mixed and “None” structures resulting from the overlap between cascades and SIA clusters is divided and added to those of $1/2\langle 111 \rangle$, $\langle 100 \rangle$ loops and C15 clusters. The allocation was made according to the relative transformation fractions of loops and C15 clusters, as shown in Fig. 4(a). The transformation fractions f_1 – f_5 in Fig. 4(a), i.e., the transformation probabilities (e.g. $f_{111 \rightarrow 100}$) in Eq. (9), adopted the MD simulation results (see Fig. 16 in Ref. [33]). Linear interpolation of available data was applied to estimate the probability values of clusters that were not covered in the MD simulations.

A key factor controlling the effect of the overlap events is F_{ol} , which gauges the probability of a cluster being fully overlapped by a cascade in unit time. The calculation for an exact value of F_{ol} requires detailed cascade simulations for a wide range of cascade energy to estimate the number density and size of (sub)cascades. Here we roughly estimate F_{ol} by relating the experimental dose rate to the volume of cascade-induced molten domains. F_{ol} is then calculated as the volume fraction of molten domains in the damage region. An early work by Backer using binary collision approximation (BCA) and MD simulation of irradiation (sub)cascades in Fe showed that the total volume $V(E)$ of molten domains caused by a primary-knock-on atom (PKA) is nearly proportional to the PKA energy E , following a fitted form $V(E) = V_{fr}(E/E_{fr})$, where V_{fr} is the molten volume at subcascade (cascade fragmentation) threshold PKA energy E_{fr} (see Fig. 3 in Ref. [67]). In alpha-Fe, E_{fr} is about 20 keV [68–70], and the corresponding V_{fr} is $\sim 1 \times 10^5$ Å³ [67]. Using the BCA code SRIM [62], the energy of Fe recoils transferred by a 150 keV incident Fe⁺ ion is calculated as a function of depth from the irradiation surface (Fig. 4(b)). Combining the PKA spectrum and the above relation between PKA energy and molten volume, the depth dependence of F_{ol} can be calculated for a given dose rate. As the model does not consider the depth effects, F_{ol} is thus averaged over a depth range where TEM observation was performed experimentally. At 300 °C, the observation of TEM foils was done at depths of 60–100 nm [55], leading to an averaged $F_{ol} = (8.34 \pm 5.80) \times 10^{-3}$ /s. At 400 °C, TEM foils at depths of 30–40 nm was investigated [22], where the averaged F_{ol} is $(8.11 \pm 0.29) \times 10^{-3}$ /s.

3.1.3. Parameters for the collapse of C15 clusters

The spontaneous collapse of C15 clusters happens when they grow exceeding critical sizes. Their collapse in the model is defined by two parameters: 1) critical collapse size of C15 clusters, and 2) the fraction of $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ loop nuclei formed via the collapse. According to MD simulations [25,27], however, the size at which C15 clusters collapse is highly stochastic. Moreover, the fraction of each loop type formed via the collapse is shown to strongly depend on the potential used for iron [27]. These uncertainties complicate the implementation of the transformation course.

In the current model, the collapse is assumed to occur when the C15 clusters contain more than 40 SIAs, which is the crossover

Table 2

Model input parameters related to irradiation condition and intrinsic property of pure alpha-Fe.

Parameters	Value
dpa rate, G_{dpa}	7.0×10^{-4} dpa/s (<i>ex-situ</i> irradiation at 300 °C) [55] 5.2×10^{-4} dpa/s (<i>in-situ</i> irradiation at 400 °C) [22,43]
Damage efficiency, η	0.3
Lattice parameter, a_0	2.842 Å
Grain size, d	30 μm *
Dislocation density, ρ_{dl}	1.0×10^8 /cm ² [63]
Dislocation bias, Z	1.1 [42]
Dislocation capture radius, r_{dc}	0.65 nm
Trap density, ρ_t	200 ppm †
Trap radius, r_t	1.5b (b: Burgers vector) [42]
Vacancy formation energy, E_f^V	1.53 eV [64]
SIA formation energy, E_f^{SIA}	3.77 eV [46]
Surface energy, γ	1.98 Jm ⁻² [65]

* The grain size of pure iron used in the experiments was several tens of microns as reported in Ref. [43]. This work adopted the value of 30 μm, as other values in the mentioned size range do not significantly modify simulation results.

† The value of trap density 200 ppm was adopted following the work of Kohnert and Wirth [42]. This value is larger than the reported content of impurities (~ 1 ppm C, <5 ppm N, <10 ppm Si and small quantities of other impurities) in pure iron [43]. We keep this value to account for the possible trapping of dislocation loops by all other trace impurities like H and O. Effects of trap density on loop evolution can be found in Ref. [42].

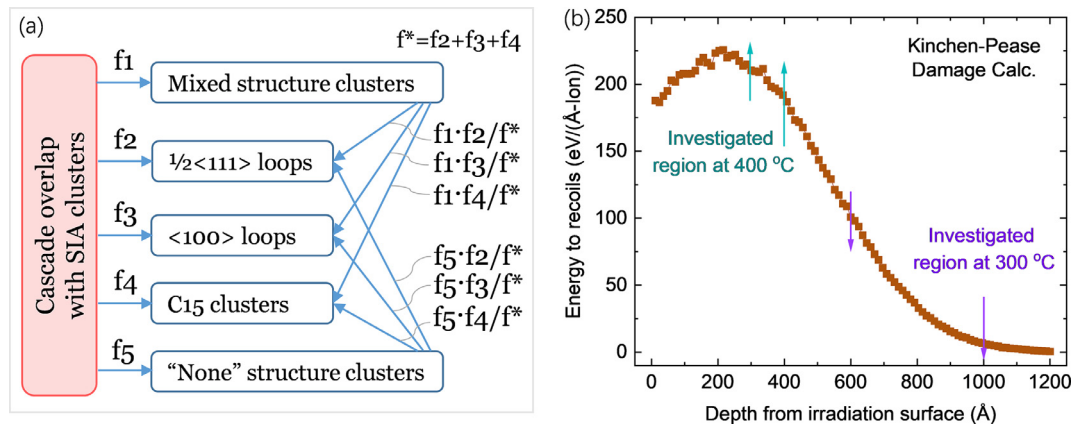


Fig. 4. Implementation of the cascade overlap with pre-existing SIA clusters in the model. (a) Various types of clusters resulting from cascade overlap with SIA clusters and their corresponding transformation fractions (f1–f5). The fractions of mixed and “None” structure clusters are divided and added to those of loops and C15 clusters. (b) PKA spectrum caused by 150 keV incident Fe⁺ ion in the pure iron target, simulated using SRIM software under the quick (K-P) mode with 100,000 ions and a displacement threshold energy of 40 eV. Note the unit on the vertical axis, which after the multiplication by dose rate gives the recoil energy per unit volume and time.

between formation energies of dislocation loops (1/2<111> type) and C15 clusters predicted by a recent atomic simulation using machine learning potentials [28]. To simplify later interpretations, this critical size is set identical at 300 and 400 °C. Over this critical size, C15 clusters collapse into loops of both types with different possibilities. The simulation using ABOP potential predicted the collapse of C15 clusters at sizes over 40 SIAs, and the collapse generated 1/2<111> and <100> loop nuclei with a fraction of 0.87/0.03. It is worth noting that the ABOP potential predicted smaller crossovers (~ 27–40 SIAs) between loops (Fig. 2 in Ref. [27]). Since the predicted size range of unstable C15 clusters (over 40 SIAs) still agrees with the estimations using ab-initial methods, we adopted the fraction of loop nuclei (0.87/0.03) in the current model.

Note that the remaining 10% of C15 clusters did not finish the collapse and were transformed into mixed loop-C15 structures. Similar to the treatment for mixed structures formed in the cascade overlap with SIA clusters, the population of mixed structures resulting from C15 clusters is allocated to 1/2<111> and <100> loops by a weight identical to their relative production rate (0.87/0.03), i.e., a fraction of 0.87/(0.87+0.03) for 1/2<111> loops and the remaining for <100> loops. The final ratio of two types of loop nuclei formed via the collapse thus becomes 0.967/0.033. At 400 °C, the fraction of <100> loops formed via the collapse may be larger, as more <100> loops (containing up to 80 SIAs) have lower energies than their 1/2<111> counterparts around the critical collapse size (Fig. 2). However, an exact value for 400 °C is unknown. To ease the later interpretation, we performed the simulation at 400 °C using the same ratio. Note that these ratio values may not exactly represent the real situation. Effects of other choices of collapse parameters will be examined in Section 3.3.

3.1.4. Comparison between simulation and experimental results

Fig. 5 shows the comparison of loop size distribution at 300 °C. The simulation overpredicted the loop density at low doses (nominal 0.65 and 1.3 dpa), while the results roughly match the experimental data for large loops at higher doses. No attempt was made to reach a better agreement, which technically can be obtained by adjusting some parameters with high uncertainty, like the assumptions on mixed and “None” structures. Interestingly, the simulation predicted a bimodal distribution of loops at high doses. The tendency of such distribution was detected experimentally in Ref. [55]. Qualitatively, the model perfectly reproduced the overall trend of loops growing with dose. Total loop density was predicted to decline at high doses (from 6.5 to 13 dpa), and the density re-

duction was significant for small loops. Large loops seem to grow at the cost of small ones by coalescence, consistent with the experimental observations [55].

For the irradiation at 400 °C, experimental loop size distribution was extracted from TEM images taken at 1.3 dpa (Fig. 2 in Ref. [22]). Total 178 loops were counted, among which 26% (46 loops) were 1/2<111> type and 74% were <100> type. Mean loop sizes were 16.1 and 6.9 nm, respectively. These re-assessed data roughly agree with those obtained in Ref. [22]. Fig. 6 shows the comparison of size distribution of loops with different Burgers vectors. Compared to 1/2<111> loops, <100> loops have a higher density while smaller mean sizes. The model reproduced this trend rather well. Quantitatively, the prediction was also satisfactory. Note that TEM observation at 400 °C was conducted for an irradiated thin-foil. Free surfaces of the thin-foil are important defect sinks for mobile defects. In the following section, we will show that the loop loss at surfaces strongly depends on the thin-foil orientation.

The comparison between the simulation and experiment, in terms of loop size distributions for two irradiation cases, may justify the validity of model, though quantitative match was not reached perfectly. Under this basis, in the following, we performed extensively the simulation using different sets of parameters. The effect of cascade overlap with pre-existing SIA clusters was explored in Section 3.2. The effect of C15 cluster stability on loop evolutions was examined in Section 3.3. The impact of loop 1-d movement, in aspects of the long-range diffusion-induced loop reactions and defect loss at surface of thin-foil in the case of *in-situ* experiments, was also addressed (Section 3.4).

3.2. Effect of cascade overlap with SIA clusters

Cascade overlap with SIA clusters provides a path for the transformation between different SIA structures. Fig. 7 shows that the density of <100> loops is significantly increased by the effects of cascade overlap with SIA clusters (COSC). For instance, the density of <100> loops smaller than ~ 7 nm (the first two bars in Fig. 7) increases from $(6.8, 6.5) \times 10^{15}$ to $(9.8, 9.3) \times 10^{15} \text{ cm}^{-3}$. The density variation for larger loops is small. The total density increase of <100> loops is about 38%. The impact on 1/2<111> loops is the opposite. Density of 1/2<111> loops decreases for loop sizes below ~ 21 nm. The total density decrease of 1/2<111> loops is about 17%. The calculation of the total number of SIAs in two types of loops shows that, the number of SIAs in 1/2<111> loops decreases from 3.12×10^{19} to $3.00 \times 10^{19} \text{ cm}^{-3}$ (–3.8%) by COSC. While COSC increases the number of SIAs in <100> loops from

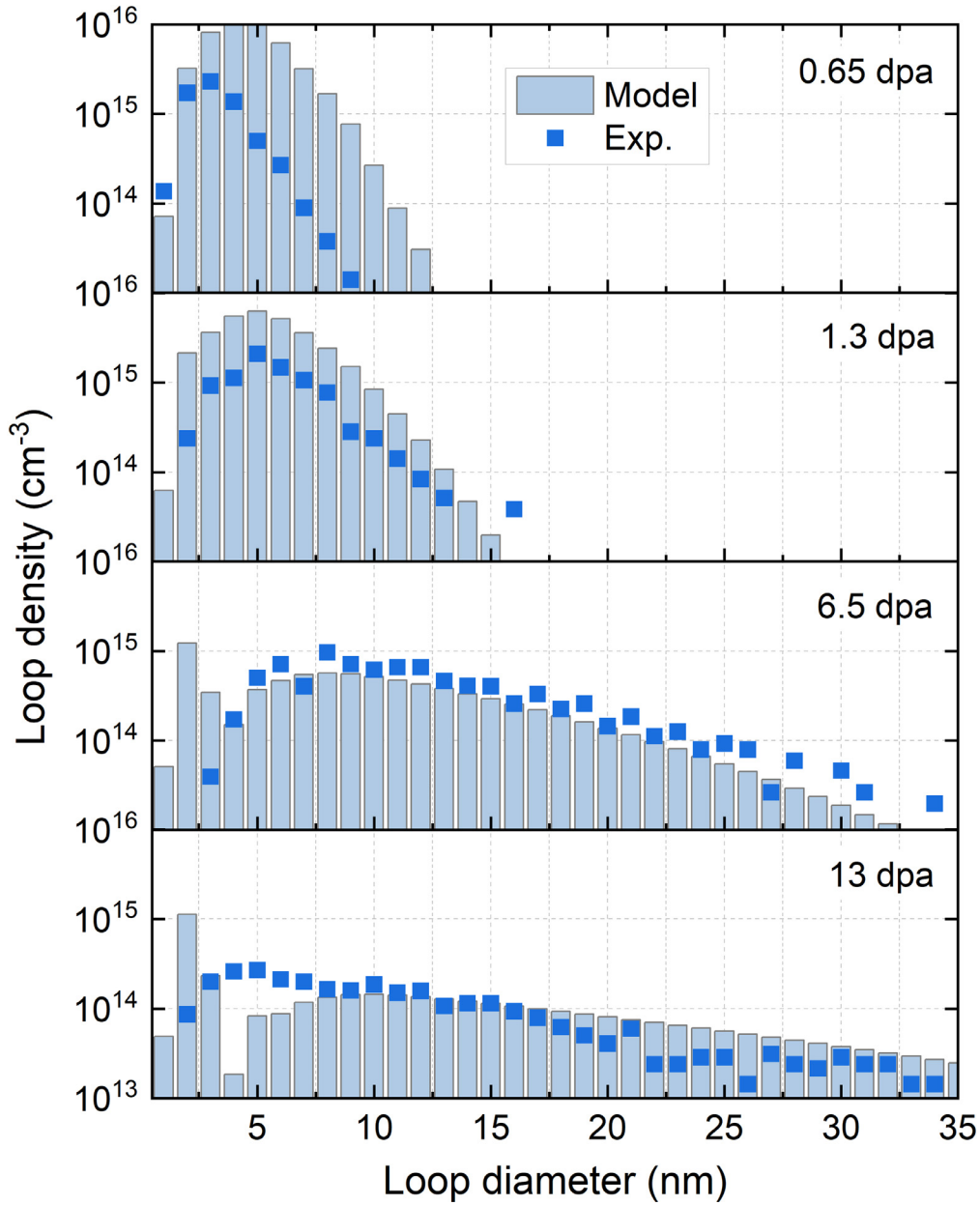


Fig. 5. Size distributions of $1/2\langle 111 \rangle$ loops simulated by the model, and probed by *ex-situ* TEM after irradiation for various dpa at 300 °C [55]. The experimental size distribution was recalculated based on the fractions of various loop sizes and total loop density, given in Yao et al. [55]. Note that a few small $\langle 100 \rangle$ loops (loop nature was unclear) were also detected. Therefore the experimental $1/2\langle 111 \rangle$ loop density at small sizes was slightly overestimated.

1.15×10^{19} to $1.44 \times 10^{19} \text{ cm}^{-3}$ (+25.2%). The number of SIAs in loops is calculated by multiplying loop density by the number of SIAs the loop contains, so the unit is still cm^{-3} . The overall effect of COSC is that some fractions of $1/2\langle 111 \rangle$ loops were reformed by the cascades into $\langle 100 \rangle$ loops. It is expected that more $\langle 100 \rangle$ loops will generate at a higher dose rate, since it enhances the probability of a cluster being overlapped by a cascade.

One may notice, however, that the increased amount of SIAs in $\langle 100 \rangle$ loops ($2.9 \times 10^{18} \text{ cm}^{-3}$) is more than the decreased amount in $1/2\langle 111 \rangle$ loops ($1.2 \times 10^{18} \text{ cm}^{-3}$). Note that the cascade overlap with C15 clusters always preferentially reforms C15 clusters into $1/2\langle 111 \rangle$ loops (see Fig. 16 of Ref. [33]). Therefore, the additional amount of SIAs gained by $\langle 100 \rangle$ loops should come from migrating SIAs. This happens because COSC causes an enlargement of $\langle 100 \rangle$ loop population and, simultaneously, a reduction of $1/2\langle 111 \rangle$ loop population in the material sys-

tem. This enhances the absorption of migrating SIAs by $\langle 100 \rangle$ loops.

3.3. Effect of C15 cluster stability

Two sets of parameters related to C15 clusters' stability were implemented in the model to explore the effect of their stability on loop evolution. The first comparison was performed between simulations assuming that C15 clusters collapse at different critical sizes. Figs. 8(a–c) show the results obtained assuming (set I) C15 clusters collapse at the critical size, $n_c = 40$ SIAs, and the ratio of $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ loops resulting from the collapse, $R_f = 0.966/0.033$; (set II) $n_c = 55$ SIAs with the same R_f . Enlarging the critical collapse size, essentially, delays the collapse of C15 clusters allowing their stability at larger sizes. A larger critical size is assumed here since there is an argument that once the cluster

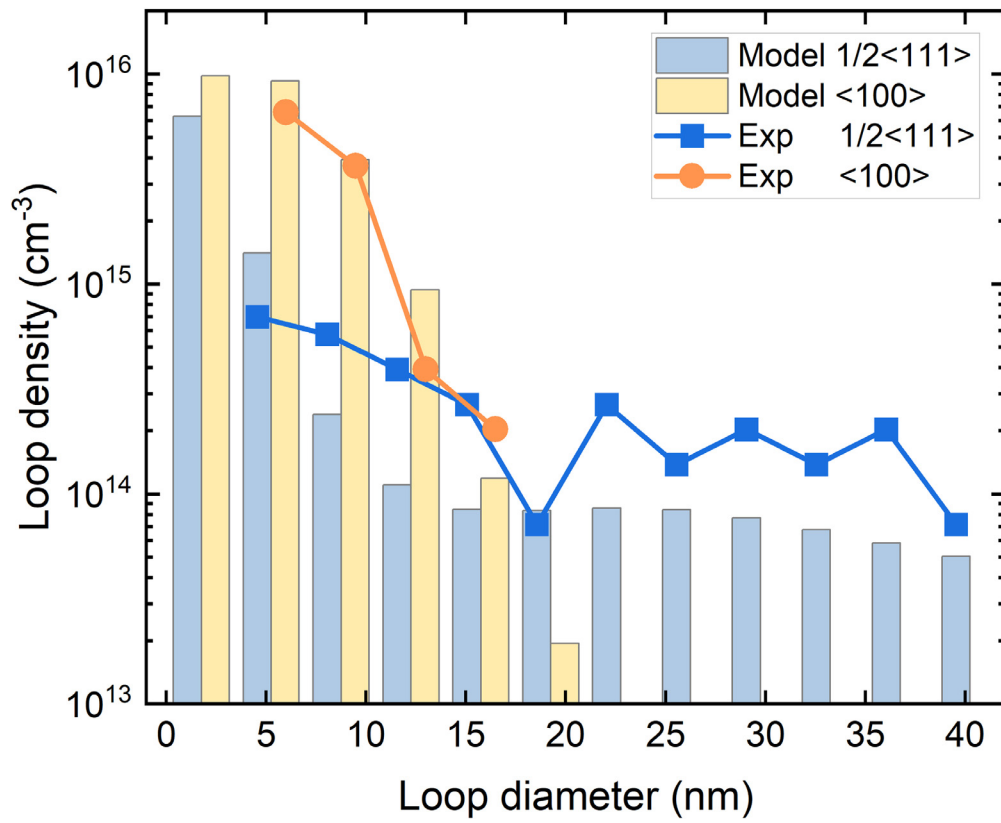


Fig. 6. Size distributions of $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ loops simulated by the model, and observed by the *in-situ* TEM at 400 °C when irradiation damage reaches 1.3 dpa. The bin width for loop statistics is 3.5 nm.

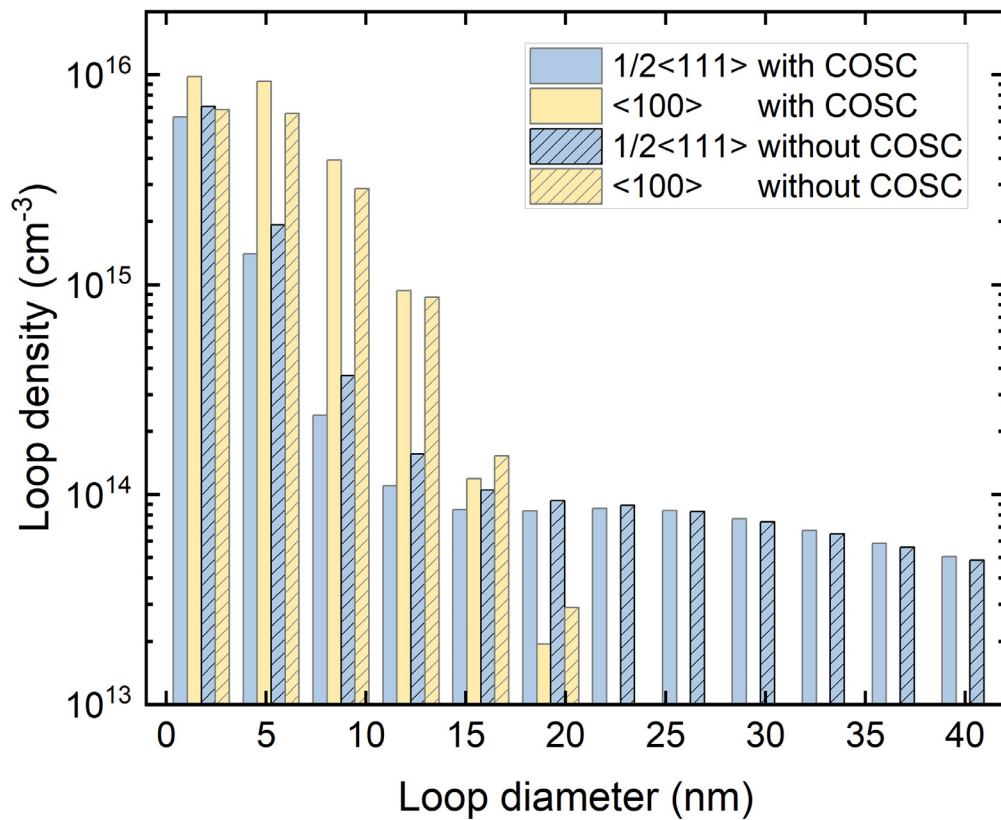


Fig. 7. Comparison of loop size distributions obtained in the simulation with and without the effects of the cascade overlap with SIA clusters (COSC). The simulation is performed for the irradiation at 400 °C and 1.3 dpa.

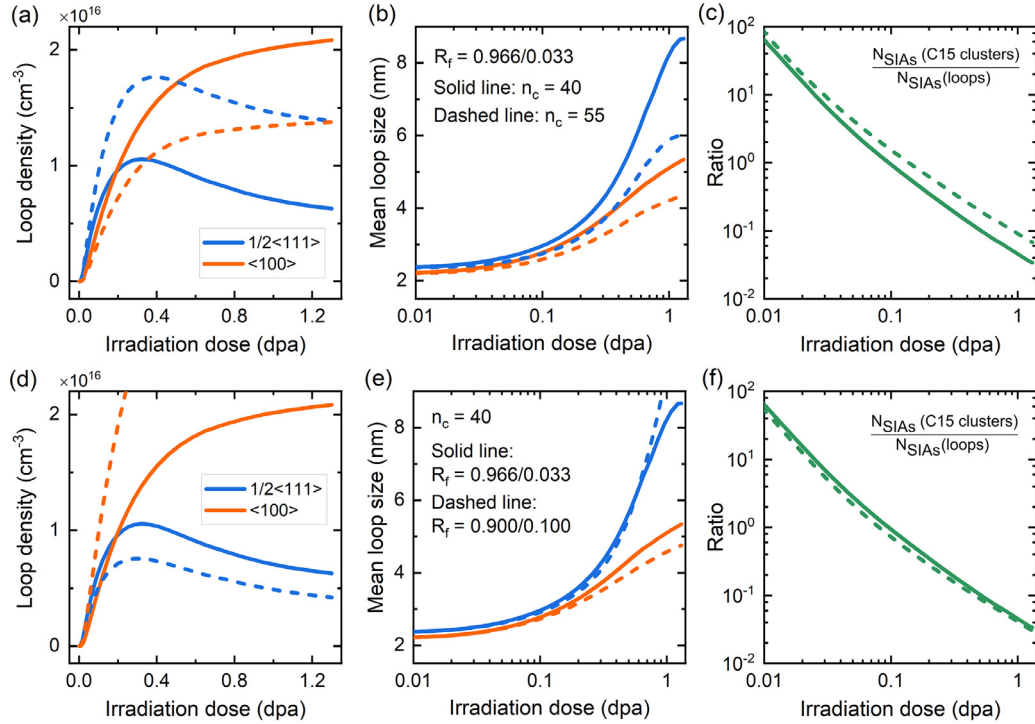


Fig. 8. Effects of C15 cluster stability on loop evolution. Comparisons of loop density, mean size and the ratio of the number of SIAs accommodated in C15 clusters and loops were shown based on simulations using two sets of parameters. Loops with size over 2 nm enter the statistics. (a–c) Simulations performed with a fixed $R_f = 0.966/0.033$, while $n_c = 40$ (solid lines) and 55 (dashed lines). (d–f) Simulations performed with a fixed $n_c = 40$, while $R_f = 0.966/0.033$ (solid lines) and 0.900/0.100 (dashed lines).

starts growing in the three-dimensional C15 morphology, it may not easily convert into the two-dimensional loop shape exactly at the energy crossover sizes [23]. Hysteresis of the collapse is expected. When $n_c = 40$, $1/2\langle 111 \rangle$ loop density increases and peaks at ~ 0.37 dpa, then decreases until 1.3 dpa. While $\langle 100 \rangle$ loops keep increasing in density. The density peak of $1/2\langle 111 \rangle$ loops was present because of the massive loss of $1/2\langle 111 \rangle$ loops caused by the four types of loop reactions shown in Fig. 1, which will be discussed later in detail. At $n_c = 55$, the delayed C15 cluster collapse lowers the density of $\langle 100 \rangle$ loops, while it remarkably increases $1/2\langle 111 \rangle$ loop density. The mean size of both loop types declines due to the delayed collapse (Fig. 8(b)). As C15 clusters are sessile structures, delaying their collapses indicates more SIAs will be trapped by them. The ratio of the number of SIAs in C15 clusters to that in loops increases (Fig. 8(c)).

If fixing the critical collapse size at $n_c = 40$, adjusting the parameter of loop nuclei ratio (R_f) modifies loop evolution in another manner. Figs. 8(d–f) show the comparison of simulations using $R_f = 0.966/0.033$ and 0.900/0.100. One can see that increasing the fraction of $\langle 100 \rangle$ loop nuclei significantly increases the density of $\langle 100 \rangle$ loops (Fig. 8(d)). Density of $1/2\langle 111 \rangle$ loops decreases. Moreover, the increased amount of $\langle 100 \rangle$ loops was larger than the decreased amount of $1/2\langle 111 \rangle$ loops. This indicates that a higher fraction of $\langle 100 \rangle$ loop nuclei via the collapse increases the total loop density. The impact on the mean size of two types of loops (Fig. 8(e)) seems to be opposite to that on their densities (Fig. 8(d)). Specifically, a larger fraction of $\langle 100 \rangle$ loop nuclei decreases the mean size of $\langle 100 \rangle$ loops, while it increases the mean size of $1/2\langle 111 \rangle$ loops at high doses over ~ 0.6 dpa. In Fig. 8(f), the ratio of the number of SIAs contained in C15 clusters and loops does not show obvious variations.

We also monitored the variation of the number of SIAs (N_{SIAs}) accommodated in loops of two types as a function of irradiation dose (Fig. 9). The critical collapse size seems to have little effect on N_{SIAs} of $1/2\langle 111 \rangle$ loops (Fig. 9(a)). The impact on that

of $\langle 100 \rangle$ loops is clear. A larger collapse size ($n_c = 55$) reduces N_{SIAs} of $\langle 100 \rangle$ loops. Therefore, the ratio of N_{SIAs} in $1/2\langle 111 \rangle$ loops to that in $\langle 100 \rangle$ loops increases compared to the simulation using $n_c = 40$. Interestingly, in both cases ($n_c = 40$ and 55), the ratio decreases from 0.01 to 0.3–0.4 dpa and then it goes up at higher doses. We recall that $\langle 100 \rangle$ loops cannot generate via the in-cascade clustering. The decreasing phase can be interpreted by the progressive nucleation and growth of $\langle 100 \rangle$ loops, e.g., via the collapse of C15 clusters, in the early stage of irradiation. At high doses over ~ 0.4 dpa, the accommodation of SIAs in $1/2\langle 111 \rangle$ loops becomes faster than that in $\langle 100 \rangle$ loops, leading to an increase of the ratio. This can be caused, for instance, by the enhanced absorption of SIAs and $\langle 100 \rangle$ loops by $1/2\langle 111 \rangle$ loops since $1/2\langle 111 \rangle$ loops grow to be much larger than $\langle 100 \rangle$ loops at high doses (Fig. 8(b)). Fig. 9(b) shows the effect of a higher fraction of $\langle 100 \rangle$ loops ($R_f = 0.900/0.100$) on N_{SIAs} in loops of both types and the corresponding ratio. Evidently, the impact on $\langle 100 \rangle$ loops is larger than $1/2\langle 111 \rangle$ loops. As expected, a higher fraction of $\langle 100 \rangle$ loop nuclei increases the accommodation of SIAs in $\langle 100 \rangle$ loops and leads to a decrease of the ratio.

3.4. Effect of defect one-dimensional diffusivity

C15 clusters generate loop nuclei over 40 SIAs, playing roles mainly at the early stage of loop evolution. Loop further evolution, however, was governed by loop interactions driven by defect long-range diffusion. In Section 3.4.1, we explored the role of the reaction between $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ loops. Additionally, for *in-situ* irradiation experiments, the free surfaces of thin-foil provide strong sinks for mobile defects. The effects of such sink sources on loop populations were addressed in Section 3.4.2.

3.4.1. Reaction between $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ loops

Fig. 10(a) shows the comparison of loop size distributions at 1.3 dpa obtained in simulations including and excluding the

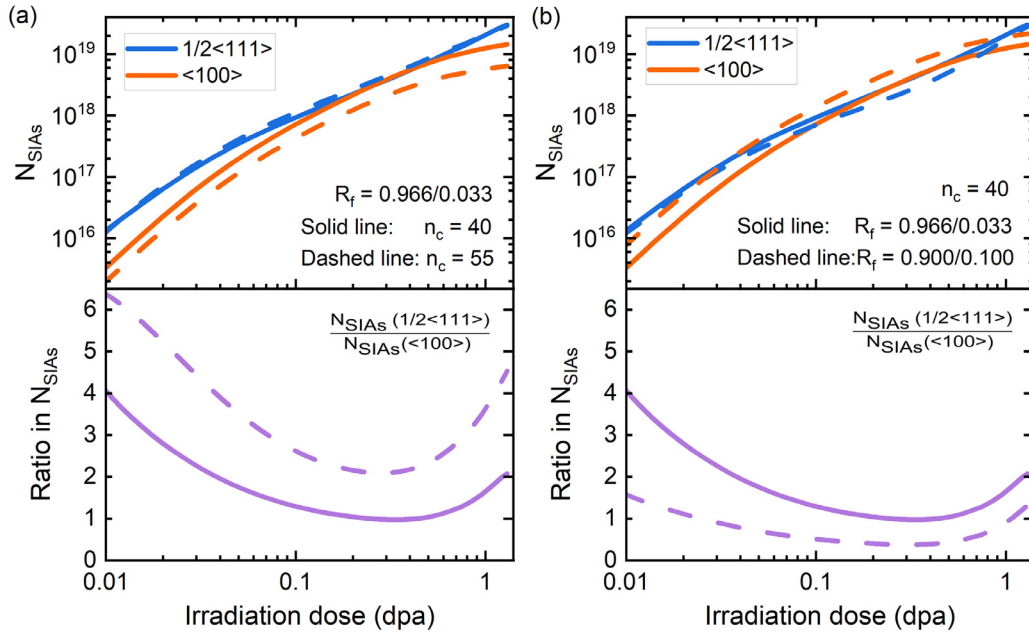


Fig. 9. Number of SIAs accommodated in $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ loops and the corresponding ratio in the number of SIAs, plotted as a function of irradiation dose. Loops with size over 2 nm enter the statistics. Comparison was made between simulations using (a) different critical collapse sizes n_c , and (b) different loop nuclei ratios R_f .

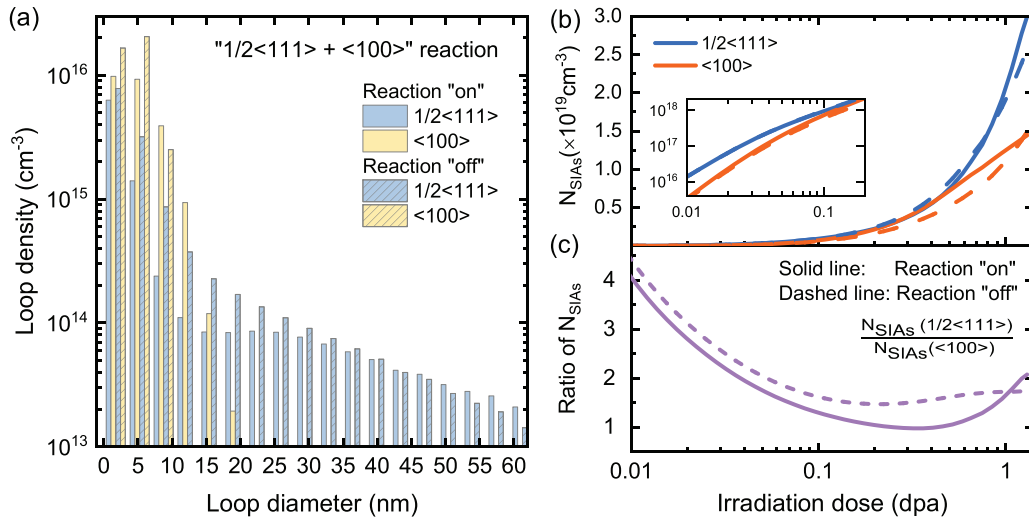


Fig. 10. Effect of the reaction between $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ loops on evolution of each loop type under irradiation at 400 °C. (a) Loop size distributions simulated at 1.3 dpa. (b) Number of SIAs accommodated in two loop types and (c) the corresponding ratio, plotted as a function of irradiation dose. The inset shows the variation of number of SIAs in a log-scale at low doses below 0.2 dpa.

" $1/2\langle 111 \rangle + \langle 100 \rangle$ " reaction. With the reaction "off", a higher density of small $\langle 100 \rangle$ loops (below ~ 7 nm) was present. Loop growth to larger ones was significantly impeded. The absence of the reaction increases the density of $1/2\langle 111 \rangle$ loops with sizes up to 40 nm. Density of $1/2\langle 111 \rangle$ loops larger than 40 nm shows a decreasing trend. Therefore, the growth of $1/2\langle 111 \rangle$ loops is also weakened by the reaction "off". These variations indicate that the " $1/2\langle 111 \rangle + \langle 100 \rangle$ " reaction essentially promotes the growth of both loops.

Fig. 10(b) shows the effect of the reaction on N_{SIAs} in loops of both types. The reaction imposes no obvious influences on N_{SIAs} of both loops before ~ 0.2 dpa. A slight increase of N_{SIAs} for $\langle 100 \rangle$ loops can be read from the inset of Fig. 10(b) for the case of reaction "on", and thus the corresponding ratio of N_{SIAs} decreases in Fig. 10(c). This reflects that the reaction plays a role in con-

tributing to the accumulation of SIAs in $\langle 100 \rangle$ loops, possibly via transferring $1/2\langle 111 \rangle$ loops into $\langle 100 \rangle$ loops. Since the amount of transferred $1/2\langle 111 \rangle$ loops is small, the variation in N_{SIAs} of $1/2\langle 111 \rangle$ loops is negligible. At 0.2–0.8 dpa, the decrease in N_{SIAs} of $1/2\langle 111 \rangle$ loops becomes observable, suggesting the increasing transfer from $1/2\langle 111 \rangle$ to $\langle 100 \rangle$ loops. Consequently, this notably rises the N_{SIAs} in $\langle 100 \rangle$ loops. At doses over 0.8 dpa, interestingly, N_{SIAs} in $1/2\langle 111 \rangle$ loops rapidly increases to be larger than that obtained with the reaction "off". This is because $1/2\langle 111 \rangle$ loops grow faster than $\langle 100 \rangle$ loops. Larger $1/2\langle 111 \rangle$ loops subsuming smaller $\langle 100 \rangle$ loops becomes significant. It thus also leads to a slowed accumulation of SIAs in $\langle 100 \rangle$ loops. These variations affect the ratio of N_{SIAs} shown in Fig. 10(c). The increasing phase of the ratio becomes insignificant when the reaction is "off". It is worth noting that the variation of N_{SIAs} is not caused by the SIA

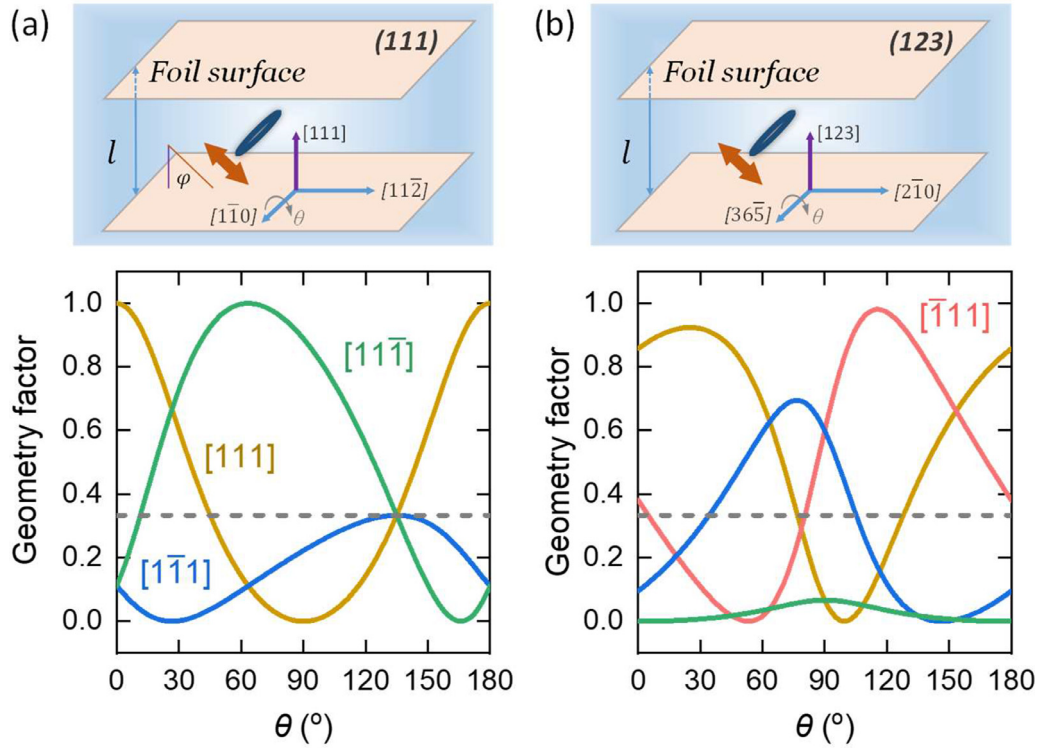


Fig. 11. Geometry factor (G_f) of loop loss at surface for four possible loop variants, plotted as a function of rotation angle (θ) with respect to surface normal. (a) Surface normal parallel to $[111]$ direction and rotates round $[1\bar{1}0]$ axis. The curves for loop variants $[111]$ and $[1\bar{1}\bar{1}]$ are perfectly overlapped. (b) Surface normal parallel to $[123]$ direction and rotates round $[36\bar{5}]$ axis. The dashed lines in both figures indicate $G_f = 0.33$, obtained by averaging G_f over four variants assuming they share an identical proportion of concentration.

transfer between loops only. The variation in loop trapping migrating SIAs also contributes to it, since the reaction modifies loop populations of both types.

In a special case where two types of loops contain an identical number of SIAs, the resultant loop is assumed to be $\langle 100 \rangle$ type in the model. The simulation assuming the resultant loop to be $1/2\langle 111 \rangle$ loops generates nearly identical results. This is due to the low possibility of such reaction events compared to all loop reactions between two types of loops.

It is evident that, loop populations after their nucleation via the in-cascade clustering (for $1/2\langle 111 \rangle$ loops only) and collapse of C15 clusters are under significant re-arrangement regulated by the loop interaction between two different types.

3.4.2. Loss of $1/2\langle 111 \rangle$ loops at free surfaces of thin-foil

Loops if mobile perform 1-d movement along directions depending on loop variants. The loss of loops due to their movement to free surfaces, therefore, will be subject to the loop movement directions and surface orientation. The current model adopted the formulas of sink strength, Eq. (7.3), developed by Huang and Mariani [49]. Fig. 11 shows the variation of the geometry factor $G_f = \cos^2\varphi$ for the four variants of $1/2\langle 111 \rangle$ loops when the surface normal rotates around a specific axis. A particular surface with $[111]$ normal and an ordinary surface with $[123]$ normal were assumed. It is clear that in most cases the sink strengths for the four loop variants are different. This suggests that $1/2\langle 111 \rangle$ loops would not distribute the concentration uniformly among the four variants. For instance, in the case shown by Fig. 11(b), $[11\bar{1}]$ variants most probably have a bigger population over other variants. These predictions were justified by the TEM characterization on loops using different diffraction vectors [22].

The current CD model cannot implement dedicated surface sink strengths for each variant. If using an averaged sink strength

$G_f = 0.33$ for the four loop variants, it significantly overestimates the real loop loss. This is because $1/2\langle 111 \rangle$ loops are highly mobile, even a small G_f leads to a great loop loss. Therefore, in the experiments by Yao et al. [22], the pure iron sample was cut into foils having the (110) surface, so as to obtain TEM specimens with a $[110]$ foil normal. In such foils, loops with Burgers vectors of $1/2[1\bar{1}1]$ and $1/2[11\bar{1}]$ have little or no component of Burger vectors towards the surface, so their loss to surfaces can be neglected. In the current simulation, we used $G_f = 0$ to reproduce the experimental data. To demonstrate qualitatively the effect of surface sinking on loop evolution, a small value of $G_f = 0.1$ is used, knowing that the magnitude of the sink strength might not be real for each loop variant in practice.

As shown in Fig. 12(a), surface sinking which targets mobile loops not only lowers the density of $1/2\langle 111 \rangle$ loops, but also $\langle 100 \rangle$ loops. Density reduction occurs notably for large loops. Moreover, the loss of $1/2\langle 111 \rangle$ loops expectedly causes the ratio of SIAs contained by them to decline (Fig. 12(b)). Moreover, the increasing phase of the ratio disappears. In either case of $G_f = 0$ or 0.1 , surface sinking described by Eq. (7.3) (3-d diffusers) for migrating SIA (clusters) and vacancy (clusters) is valid. Since the surface always provides strong sinks for SIA and vacancy (and their migrating clusters), the variation in loop populations does not significantly modify their densities. Therefore, the variations in loop population, shown in Fig. 12(a), were mainly caused by the loss of $1/2\langle 111 \rangle$ loops at surfaces. The density reduction of large loops of both types can be explained by decelerated loop coalescence, since the rates of which decrease with the decreasing loop populations.

4. Discussion

Previous explanations for the increasing occurrence of $\langle 100 \rangle$ loops at high temperatures employed the argument that $1/2\langle 111 \rangle$

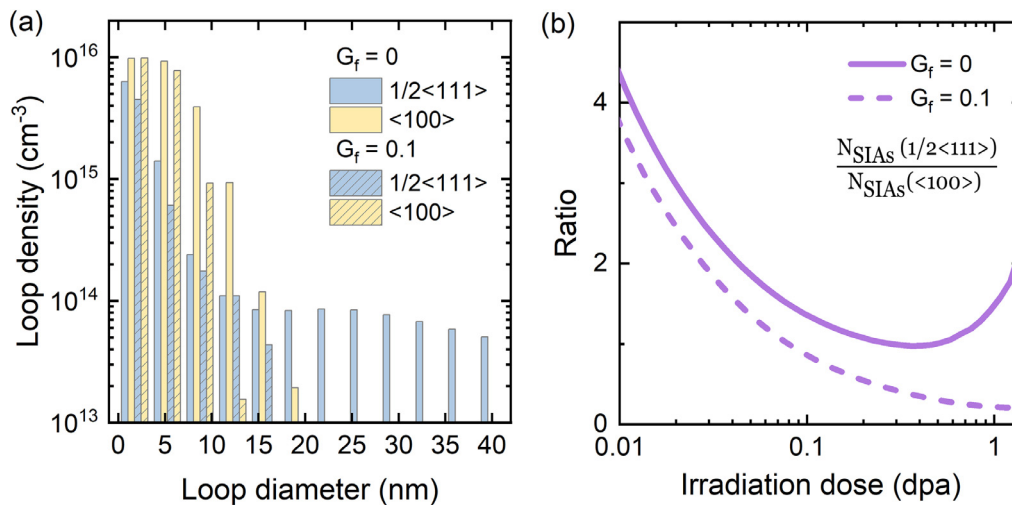


Fig. 12. Effects of surface sinking on defect evolution. Foil thickness is 50 nm. Comparison of (a) density of $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ loops, (b) ratio of the number of SIAs accommodated in loops of different types at 1.3 dpa simulated with $G_f = 0$ and 0.1 at 400 °C.

loops become energetically unstable compared to the $\langle 100 \rangle$ type at elevated temperatures [21,22]. $1/2\langle 111 \rangle$ loops if form at high temperatures will transfer into $\langle 100 \rangle$ type, as suggested by the mechanism, though it has not been justified experimentally. Our simulation results suggest that both $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ loops form in the early stage of loop evolution. The model considers three sources of $1/2\langle 111 \rangle$ loops nucleation: (1) the in-cascade clustering, (2) accumulation of migrating SIAs and (3) the collapse of C15 clusters; $\langle 100 \rangle$ loops nucleate mainly via the collapse of C15 clusters. We also consider $\langle 100 \rangle$ loop formation from the interaction between two $1/2\langle 111 \rangle$ loops of identical size. However, simulation results show that this contribution to the nucleation of $\langle 100 \rangle$ loops is negligible. This is due to the low possibility of two $1/2\langle 111 \rangle$ loops with an identical size and suitable orientations to encounter each other. During the long-range movement of $1/2\langle 111 \rangle$ loops, they can be incorporated into, or, on the contrary, absorb $\langle 100 \rangle$ loops, depending on loop relative sizes. Hence, loop relative population will be under re-arrangement caused by microstructural long-term evolutions. The population re-arrangement proceeds not because a single $1/2\langle 111 \rangle$ loop directly transfers its orientation into the $\langle 100 \rangle$ configuration. Instead, loop mutual interactions provoke the population modification, as suggested by the simulation results.

Our calculations on loop energetics imply that small loops with $\langle 100 \rangle$ Burgers vector are energetically more stable than those with $1/2\langle 111 \rangle$ configurations (Fig. 2). This prediction, in a sense, agree with the DFT predictions [24], though the energy crossover between two loop types was predicted at sizes around 8 SIAs, smaller than that predicted by the anisotropic elasticity. This may be a temperature effect since DFT results are valid at 0 K. Moreover, it was found that the stability of $\langle 100 \rangle$ loops expands to larger sizes with increasing temperatures (Fig. 2(b)). If following Byggmästar's speculation that the restructuration of collapsing C15 clusters abides by the tendency that they transform into loops with the lowest energy configurations [27], the estimated increasing stability of $\langle 100 \rangle$ loops over $1/2\langle 111 \rangle$ loops with temperature in this study suggests that C15 clusters tend to collapse into $\langle 100 \rangle$ types as temperature rises. However, this deduction needs to be justified by the energy landscape, transition mechanisms and the barriers of C15 cluster transformation into different types of loops.

The model has considered two critical sizes for the collapse of C15 clusters. The simulation using $n_c = 40$ SIAs considers that the collapse occurs exactly at the energy crossover between C15 clusters and loops. A larger critical size of $n_c = 55$ SIAs assumes

the delayed collapse, considering the hysteresis of conversion of a C15 cluster from a three-dimensional morphology to a two-dimensional one at the energy crossover. A larger critical size also means that C15 clusters can accommodate more SIAs and immobilize them. This is one of the reasons for the decrease in the loop mean size of both types at $n_c = 55$ SIAs (Fig. 8(b)). Since the collapse of C15 clusters is the major source of $\langle 100 \rangle$ loop nucleation in the model, C15 cluster collapses at larger sizes means that $\langle 100 \rangle$ nucleation will rely on density of large C15 clusters. In the model, C15 clusters grow by trapping migrating SIAs. Larger C15 clusters therefore is always less than smaller ones. Thus, a larger critical collapse size of C15 clusters decreases the density of $\langle 100 \rangle$ loops (Fig. 8(a)). Certainly, $1/2\langle 111 \rangle$ loop nuclei generated via the collapse also become less. However, nucleation of $1/2\langle 111 \rangle$ loops is mainly from the other two sources. Collapse of C15 clusters makes up a rather small fraction of $1/2\langle 111 \rangle$ loop nuclei, though most C15 clusters collapse into $1/2\langle 111 \rangle$ loops. This is because only 7% of SIA clusters hold the C15 structure via the in-cascade clustering. Therefore the variation of critical collapse size would not significantly affect the nucleation of $1/2\langle 111 \rangle$ loops.

However, the decrease in $\langle 100 \rangle$ loop population enhances the mean free path of $1/2\langle 111 \rangle$ loops (λ parameter in Eq. (4)). This happens since other components of defects and sinks interacting with gliding $1/2\langle 111 \rangle$ loops do not vary significantly using different critical sizes. Note that interaction between C15 clusters and $1/2\langle 111 \rangle$ loops was not considered in the model. A larger mean free path of 1-d diffusing loops, thus, slows the transfer reactions between $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ loops, as well as the coalescence reactions between $1/2\langle 111 \rangle$ loops. Hence, a much higher density of $1/2\langle 111 \rangle$ loops was present at $n_c = 55$ SIAs (Fig. 8(a)). The reduced reaction rates are also responsible for the smaller mean sizes of both loop types (Fig. 8(b)).

In the simulation assuming a higher fraction of $\langle 100 \rangle$ loop nuclei in the collapse of C15 clusters ($R_f = 0.900/0.100$), a significant increase of $\langle 100 \rangle$ loop density occurs as expected (Fig. 8(d)). However, a notable decrease of $1/2\langle 111 \rangle$ loop density is also present. We emphasize again that the variation in $1/2\langle 111 \rangle$ loop density does not result directly from the reduced fraction of $1/2\langle 111 \rangle$ loop nuclei generated by the collapse. Instead, it is caused by the enhanced reaction rates between loops. More $\langle 100 \rangle$ loop population generated via the collapse increase $\langle 100 \rangle$ loop population, which shortens the mean free path of diffusing $1/2\langle 111 \rangle$ loops. Consequently, this enhances the reaction rates involving $1/2\langle 111 \rangle$ loops. Therefore, at $R_f = 0.900/0.100$, density of

$1/2\langle 111 \rangle$ loops becomes lower (Fig. 8(d)), and larger mean sizes are present at high doses (Fig. 8(e)). The mean size of $\langle 100 \rangle$ loops, however, is calculated to be smaller at $R_f = 0.900/0.100$. There are several reasons for this. On the one hand, $\langle 100 \rangle$ loop population contains a larger number of small loops produced by the increased nucleation of $\langle 100 \rangle$ loops. On the other hand, growth of $\langle 100 \rangle$ loops via the path of interacting with $1/2\langle 111 \rangle$ loops requires a $\langle 100 \rangle$ loop encountering a $1/2\langle 111 \rangle$ loop smaller than it. The enhanced coalescence between $1/2\langle 111 \rangle$ loops reduces the population of small $1/2\langle 111 \rangle$ loops. Moreover, absorption of $\langle 100 \rangle$ loops by larger $1/2\langle 111 \rangle$ loops also contributes to a smaller mean size of $\langle 100 \rangle$ loops.

In the implementation of C15 cluster collapse in the model, an assumption on the mixed structures was made. The mixed structures were assumed to finally evolve into loops of both types according to the relative fraction of loop nuclei resulting from the collapse. This may overestimate the loop populations, since some of the mixed structures were shown to be quite stable under annealing at 1000 K [33]. Besides, a weight of $0.83/(0.83+0.03)$ allocated to $1/2\langle 111 \rangle$ loops essentially assume that most mixed structures evolve into $1/2\langle 111 \rangle$ loops. This may underestimate the fraction of the mixed structures evolving into $\langle 100 \rangle$ loops. These effects can be explored by further adjusting the fraction ratio of loop nuclei generated from C15 clusters.

Note that C15 cluster formation and collapse occur during the entire process of irradiation. However, when loops accumulated to a large population and size, SIAs required by the growth of C15 clusters are reduced due to the increasing trapping of SIAs by loops. Hence, the influence of C15 clusters on loop evolution weakens at high doses. This can be also read from the decreasing ratio of the number of SIAs in C15 clusters to that in loops (Figs. 8(c,f)).

Though $\langle 100 \rangle$ loop nuclei were much less than $1/2\langle 111 \rangle$ loops in all proposed setups for the collapses, density of $\langle 100 \rangle$ loops become comparable or much larger than $1/2\langle 111 \rangle$ loops after long-term evolution (Figs. 8(a,d)). One can also notice that density of $1/2\langle 111 \rangle$ loops always shows a declining trend at high doses (forming a density peak). It happens because the production of $1/2\langle 111 \rangle$ loops becomes slower than the “loss” at high doses. The “loss” in terms of density is caused by the trapping of $1/2\langle 111 \rangle$ loops by inherent sinks, $1/2\langle 111 \rangle$ loops coalescing into larger ones and $1/2\langle 111 \rangle$ loops being subsumed by larger $\langle 100 \rangle$ loops.

In the long-term loop evolutions, the reaction of “ $1/2\langle 111 \rangle + \langle 100 \rangle$ ” was identified to cause SIAs transfer between loops of different types. It was demonstrated that the reaction leads to the variation in loop relative populations (Fig. 10(a)). Note that, for instance, a $1/2\langle 111 \rangle$ loop subsuming a $\langle 100 \rangle$ loop to become a larger $1/2\langle 111 \rangle$ loop does not change the total density of $1/2\langle 111 \rangle$ loops, while it promotes loop growth at the cost of a smaller loop of another type. Therefore, the overall effect of the reaction is that it reduces the total loop population and re-arranges the SIAs accommodation in loops of two types. The reduction of loop population is easy to be understood since the “ $1/2\langle 111 \rangle + \langle 100 \rangle$ ” is still a type of coalescence reaction. The re-arrangement of SIAs, i.e., SIAs transfer between loops of different types, depends on loop relative sizes.

The rate of SIA transfer driven by a $\langle 100 \rangle$ loop containing j SIAs can be computed as

$$\text{Trans}_{100}(j) = \sum_{j>i} k_i^j C_{100}(j) C_{111}(i) \cdot i - \sum_{j<i'} k_{i'}^j C_{100}(j) C_{111}(i') \cdot j \quad (12)$$

where the first term on the right-hand side describes $\langle 100 \rangle$ loops absorbing $1/2\langle 111 \rangle$ loops of smaller sizes i and thus causing the gain of SIAs by $\langle 100 \rangle$ loops. The second term denotes $\langle 100 \rangle$ loops being absorbed by $1/2\langle 111 \rangle$ loops of larger size i' and thus causing SIAs loss of $\langle 100 \rangle$ loops. Positive $\text{Trans}_{100}(j)$ indicates

that $\langle 100 \rangle$ loops of size j gain SIAs from $1/2\langle 111 \rangle$ loops via the reaction. They lose SIAs for negative $\text{Trans}_{100}(j)$. A similar equation can be generated for $1/2\langle 111 \rangle$ loops. In the practical calculation, Eq. (12) is reformed into a new expression according to the G-grouping scheme described in Section 2.3. It is clear that the transfer rate depends on loop size and corresponding loop density which varies with irradiation dose.

Fig. 13(a) shows that $1/2\langle 111 \rangle$ loops gaining SIAs at low doses shifts to losing SIAs at high doses. This happens for loops below 8 nm. Larger loops progressively get positive Trans_{111} with the increase of dose. For $\langle 100 \rangle$ loops, Fig. 13(b) shows that small loops below ~ 7 nm always hold positive Trans_{100} up to 1.3 dpa. The rates of gaining SIAs, however, decrease with the increase of dose. At 1.3 dpa, Trans_{100} of loops with sizes of 8–16 nm decreases and becomes even negative, indicating the occurrence of SIA loss. These variations are closely correlated with loop evolutions. For instance, the increasing rates of small $1/2\langle 111 \rangle$ loops losing SIAs at higher doses are caused by the growth of $\langle 100 \rangle$ loops. Small $\langle 100 \rangle$ loops of 2–7 nm always gain SIAs because of the continuous generation of a large number of smaller $1/2\langle 111 \rangle$ loops (containing up to 16 SIAs) via in-cascade clustering and accumulation of migrating SIAs. As $1/2\langle 111 \rangle$ loops grow large at high doses, $\langle 100 \rangle$ loops losing SIAs become significant and thus negative Trans_{100} is present. Since density of $\langle 100 \rangle$ loops larger than 16 nm is very low, the transfer rates of these loops are always small. We sum up loop transfer rates, which gives the total transfer rates of the loop population larger than 2 nm as a function of dose (Fig. 13(c)). Results show that the total transfer rates are always positive for each loop type. This explains that the “ $1/2\langle 111 \rangle + \langle 100 \rangle$ ” reaction promotes the growth of both loops. Moreover, with the increase of dose, the rate of SIA transfer from $1/2\langle 111 \rangle$ to $\langle 100 \rangle$ loops declines and is lower than that from $\langle 100 \rangle$ to $1/2\langle 111 \rangle$ loops at doses over 0.6 dpa. This evidences the faster accommodation of SIAs in $1/2\langle 111 \rangle$ loops than in $\langle 100 \rangle$ loops at high doses (Fig. 10(b)). The physics behind the transfer was discussed in detail by Arakawa, interested readers can refer to Ref. [35]. Briefly, the attractive force between loop segment and junction drives the reaction.

In the case of *in-situ* irradiation experiments, $1/2\langle 111 \rangle$ loop variants with orientation close to the surface normal of thin-foils fast glide to surfaces. Resulting from the unequal loss to sinks, the density inequality among loop variants, in turn, affects the nominal reaction rates in the model. It can be expected that the possibility of reaction between aligned loops will increase, if loop loss occurs mainly on one or two particular loop variants. This can be a real case, for instance, a thin-foil with the orientation shown in Fig. 11(b) with θ around 50° or 140° . The density inequality of $1/2\langle 111 \rangle$ variants does not affect their reaction cross section with $\langle 100 \rangle$ loops, as the angle between orientations of a $\langle 100 \rangle$ loop and any variant of $1/2\langle 111 \rangle$ loops is always $\sim 54.7^\circ$. However, loss of $1/2\langle 111 \rangle$ loops still has an impact on $\langle 100 \rangle$ loops. On the one hand, the resulting reduction in loop total population reduces the general reaction rates (via affecting the mean free path of loops) between $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ loops. More importantly, loss of $1/2\langle 111 \rangle$ loops at surfaces reduces the transfer of SIAs to $\langle 100 \rangle$ loops (Fig. 12(b)). This suggests that, for *in-situ* (thin-foil) experiments, foil orientation influences populations of both loop types. Experimentally, Yao et al. [22] indeed found in foil irradiation that only loops with particular Burgers vectors grew large, while others kept small sizes during irradiation, in line with the discussions here. It is worth mentioning that large vacancy clusters or voids were not predicted in simulations, again agreeing with experimental observations [22]. Small mobile vacancy clusters formed easily moved to sinks or were annihilated by loops.

A final question concerns the temperature effects on relative population of the two loop types. In essence, temperature effects

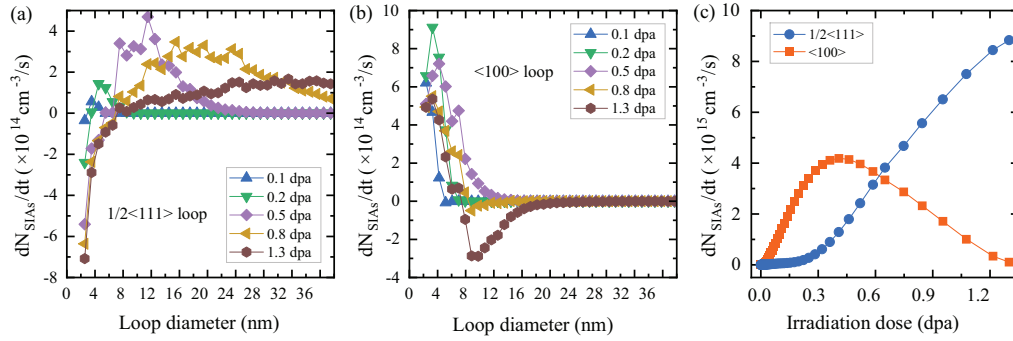


Fig. 13. SIA transfer rate of loops at various irradiation doses at 400 °C, plotted as a function of loop size (over 2 nm). (a) $1/2\langle 111 \rangle$ loop, (b) $\langle 100 \rangle$ loop. (c) Total SIA transfer rate of $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ loops larger than 2 nm.

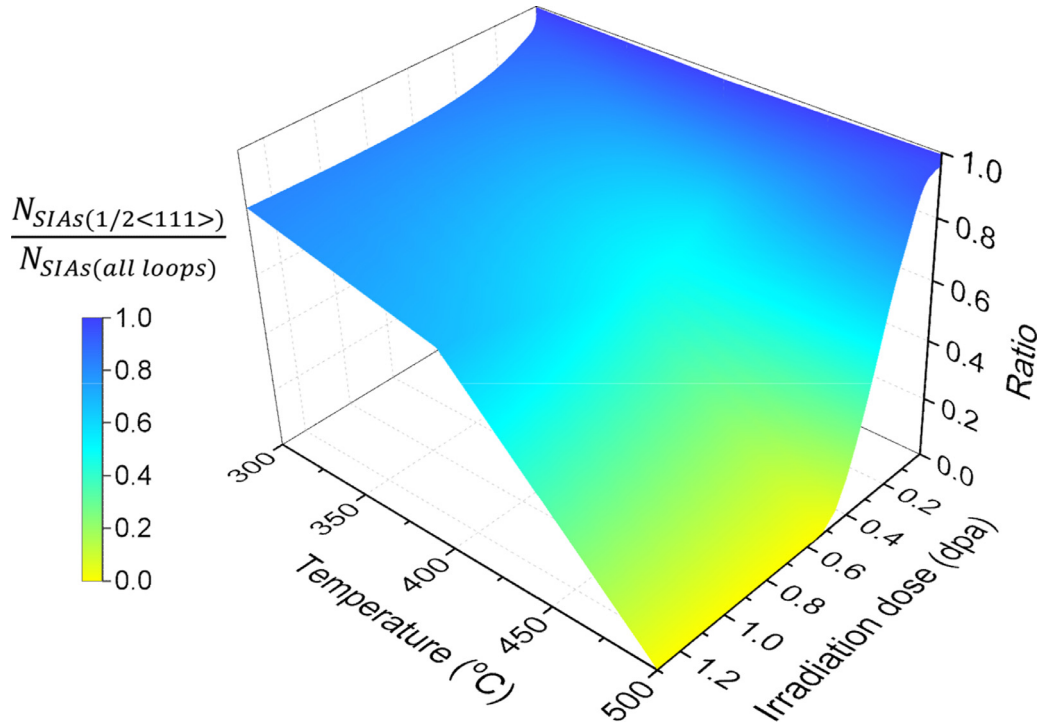


Fig. 14. Temperature and dose dependence of the ratio of number of SIAs accommodated in $1/2\langle 111 \rangle$ loops.

on C15 cluster stability are not entirely clear yet, though the loop energetics has indicated the increasing stability of $\langle 100 \rangle$ loops over $1/2\langle 111 \rangle$ loops with the increase of temperature. Note that the energetics of C15 clusters at high temperatures were not taken into consideration, which should also matter in terms of collapse sizes. Apart from the effect on C15 clusters, temperature, on the other hand, certainly influences the loop population via modifying defect diffusivities. Fig. 14 shows the fraction of SIAs contained in $1/2\langle 111 \rangle$ loops varying with irradiation dose up to 1.3 dpa at temperatures ranging from 300 to 500 °C in iron. For the simulation at 500 °C, the parameters of C15 cluster collapse adopted those used at 400 °C, knowing that this may underestimate the fraction of $\langle 100 \rangle$ loop nuclei via the collapse. The result clearly demonstrates that the SIAs transfer between two loop types was significantly promoted at high temperatures. Note that loop loss at dislocation lines and grain boundaries was also enhanced, but the contribution to the decrease of $1/2\langle 111 \rangle$ loop population was marginal compared to that caused by $\langle 100 \rangle$ loops. Even though $1/2\langle 111 \rangle$ loops occupy most of loop nuclei resulting from C15 cluster collapse at 500 °C and they can form continuously from cascades, al-

most all of them are subsumed by $\langle 100 \rangle$ loops during their long-range movements. Note that a high temperature in principle also promotes fast growth of $1/2\langle 111 \rangle$ loops. However, the density of large loops is many orders of magnitude lower than $\langle 100 \rangle$ loops in the simulation at 500 °C. This occurs because small $1/2\langle 111 \rangle$ loops once formed are quickly subsumed by sessile $\langle 100 \rangle$ loops, which significantly reduces their density. Therefore, the dominant transfer of SIA from $\langle 100 \rangle$ loops to $1/2\langle 111 \rangle$ loops at high doses, as shown in Fig. 13(c) for the simulation at 400 °C, does not occur at 500 °C. Hence, the fraction of SIA accommodated in $1/2\langle 111 \rangle$ loops at 500 °C decreases fast with irradiation dose, and reaches the value close to zero at ~ 0.5 dpa (Fig. 14). These predictions are in perfect consistency with the experimental findings that no visible $1/2\langle 111 \rangle$ loops were observed at 500 °C [22].

5. Conclusion

In the present work, we intensively use cluster dynamics simulations to investigate the evolution of $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ interstitial loops in alpha-Fe under irradiation. The kinetics of C15

clusters formation and subsequent collapse into loops of different types were implemented in the model. Transformation between SIA clusters caused by cascade overlapping with pre-existing SIA clusters was also considered. Moreover, the model adopted the 1-d reaction kinetics for interactions involving the interstitial loops. The energetics of interstitial loops of two types at high temperatures were assessed using anisotropic elasticity theory to examine the temperature dependence of loop evolutions. The role of the C15 cluster stability and defect 1-d movement in the variation of relative population of $1/2\langle 111 \rangle$ and $\langle 100 \rangle$ interstitial loops was addressed in detail. Main results can be summarized as follow:

- (1) C15 clusters essentially act as SIA buffers by accommodating SIAs and immobilizing them. The larger the collapse size, the larger the capacity of the buffers. The variation of the collapse parameters affects loop evolution of both types.
- (2) In the simulation at 400 °C without loop loss at surfaces, C15 cluster collapse at larger sizes reduces significantly the density of $\langle 100 \rangle$ loops, and decreases the reaction rates between loops, leading to a higher density of $1/2\langle 111 \rangle$ loops and smaller loops of both types in the TEM visible size range. A higher fraction of $\langle 100 \rangle$ loop nuclei via the collapse generates more $\langle 100 \rangle$ loops, and thus promotes $1/2\langle 111 \rangle$ loops interacting with $\langle 100 \rangle$ loops and themselves, leading to larger but lower density of $1/2\langle 111 \rangle$ loops and reduced mean size of $\langle 100 \rangle$ loops.
- (3) Loop energetics estimations reveal that $\langle 100 \rangle$ loops are more stable over $1/2\langle 111 \rangle$ loops with increasing temperatures. The indication on the preferred type of loop nuclei resulting from the collapse of C15 clusters, however, warrants further investigations, for instance on the energy landscape of the transformation of C15 structures into different types of loops.
- (4) Cascade overlapping with pre-existing SIA clusters indeed provides another path for the nucleation of $\langle 100 \rangle$ loops. More $\langle 100 \rangle$ loops are expected to generate through this path at higher dose rates.
- (5) The reaction " $1/2\langle 111 \rangle + \langle 100 \rangle$ " was identified as the origin causing the SIA transfer between loops of different types during loop long-term evolution. The reaction kinetically provides a path for loop growth of both types. $1/2\langle 111 \rangle$ loops transferring into components of $\langle 100 \rangle$ loops dominates at the early stage of irradiation. SIA transfer from $\langle 100 \rangle$ loops to $1/2\langle 111 \rangle$ loops depends on the evolution of large $1/2\langle 111 \rangle$ loops. Under the condition of strong loss of $1/2\langle 111 \rangle$ loops at surfaces (thin foil experiments) and/or temperature-enhanced absorption of small $1/2\langle 111 \rangle$ loops by $\langle 100 \rangle$ loops (e.g. at 500 °C), the overall SIA transfer from $\langle 100 \rangle$ loops to $1/2\langle 111 \rangle$ loops is significantly weakened.
- (6) Irradiation of thin-foils causes density inequality among $1/2\langle 111 \rangle$ loop variants. Foil orientation influences populations of both loop types.
- (7) Temperature strongly affects loop relative populations via influencing defect diffusivities. Even if allowing the majority of collapsing C15 clusters to construct into $1/2\langle 111 \rangle$ loop nuclei at high temperatures, and with continuous nucleation of $1/2\langle 111 \rangle$ loops via in-cascade clustering and the accumulation of migrating SIAs, most $1/2\langle 111 \rangle$ loops are merged into components of $\langle 100 \rangle$ loops during their long-range movements when the temperature is over (equal) 500 °C.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work is supported by the [Helmholtz Association](#) of German Research Centers (HGF) and has been carried out within the framework of Nuclear Fusion Programme at Karlsruhe Institute of Technology (KIT), Germany.

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